

Basic Router Configuration

- These presentations are the result of a collaboration among the instructors at St. Clair College in Windsor, Ontario.
- Thanks must go out to Rick Graziani of Cabrillo College. His material and additional information was used as a reference in their creation.

Configuring and Testing Your Network

Configuring Cisco Devices: IOS Basics



```
version 12.2
!
hostname Router
!
!interface FastEthernet0/0
no ip address
duplex auto
speed auto
shutdown
!
interface Serial0/0
no ip address
shutdown
!
interface Serial0/1
no ip address
shutdown
```

Cisco IOS



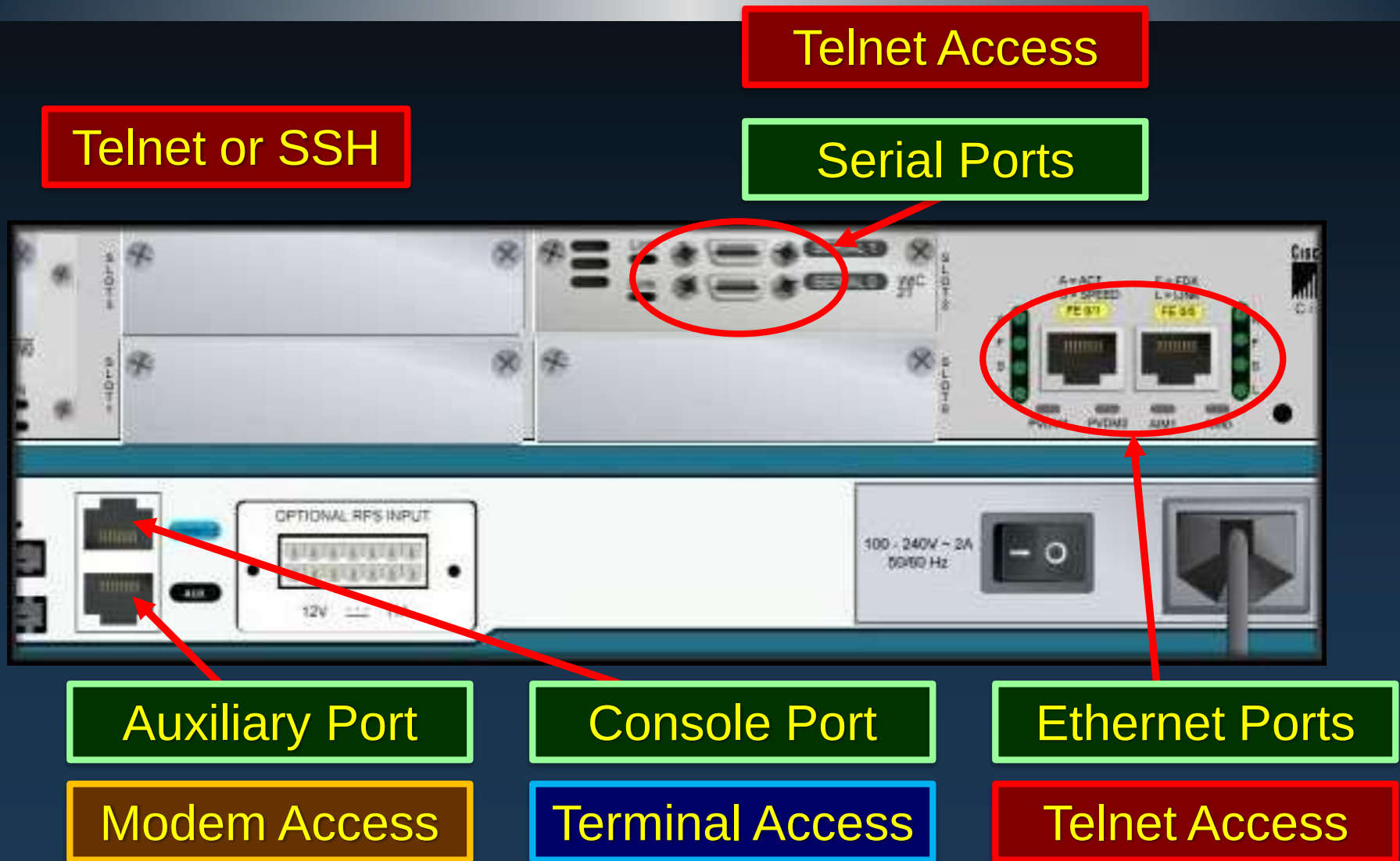
- As with a computer, a router or switch cannot function without an operating system.
- Cisco calls its operating system the **Cisco Internetwork Operating System** or **Cisco IOS**.
- The Cisco IOS provides the following network services:
 - Basic routing and switching functions
 - Reliable and secure access to networked resources
 - Network scalability

Cisco IOS



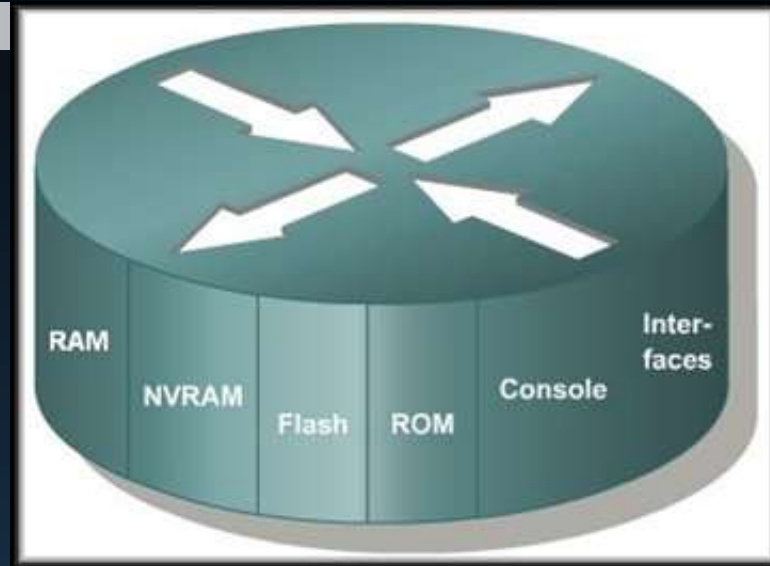
- The Cisco IOS is accessed using the Command Line Interface (CLI).
- Features will vary based on the version of the IOS and the type of device.

Access Methods



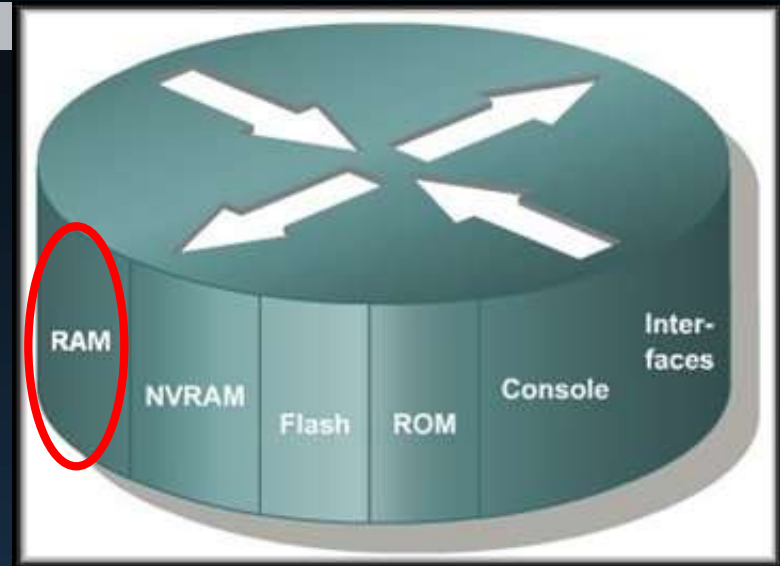
Router Components

- The main internal components of a router are:
 - Random Access Memory (RAM)
 - Nonvolatile Random Access Memory (NVRAM)
 - Flash Memory (Flash)
 - Read Only Memory (ROM)
 - Console
 - Interfaces



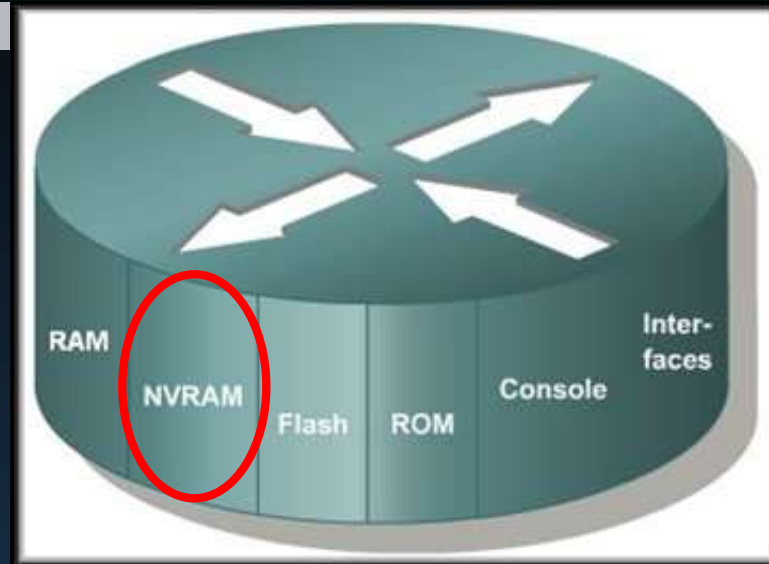
Router Components

- Random Access Memory (RAM):
 - Stores routing tables
 - Holds ARP cache
 - Holds fast-switching cache
 - Performs packet buffering as shared RAM
 - Maintains packet-hold queues
 - Provides temporary memory for the configuration file of a router while the router is powered on
 - Loses content when a router is powered down or restarted



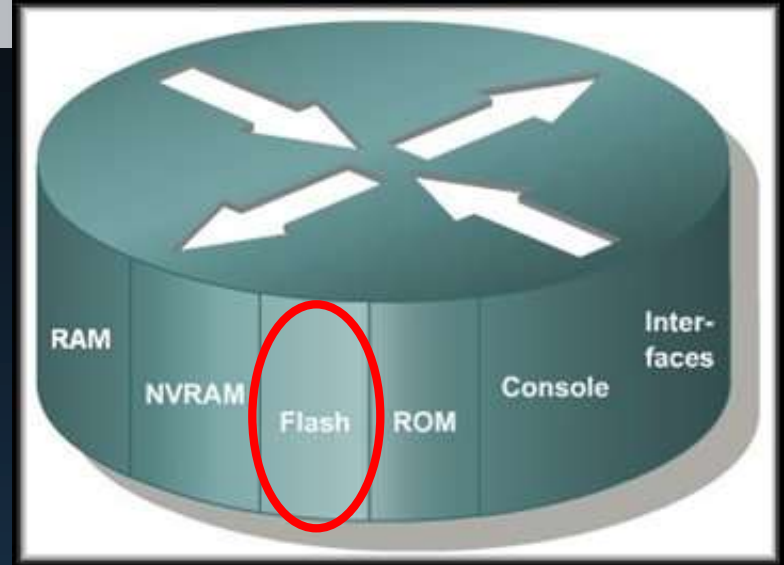
Router Components

- Nonvolatile Random Access Memory (NVRAM):
 - Provides storage for the startup configuration file
 - Retains content when a router is powered down or restarted



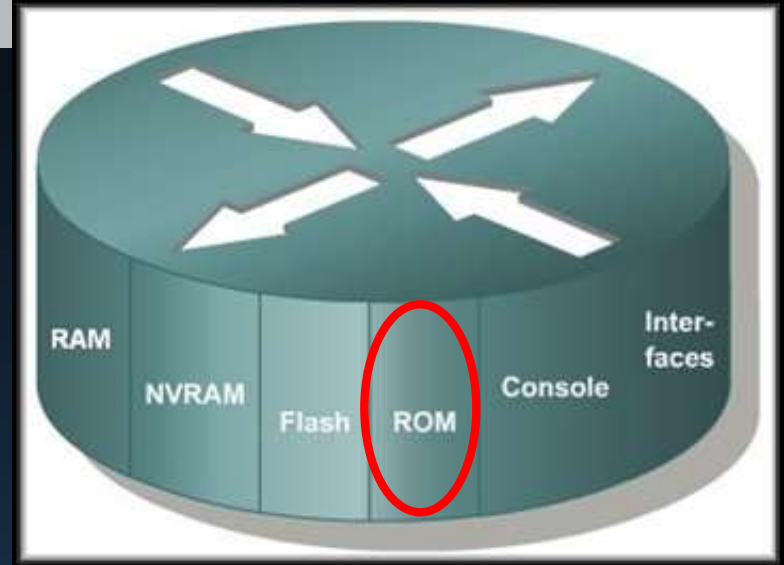
Router Components

- Flash Memory (Flash):
 - Holds the IOS image
 - Allows software to be updated without removing and replacing chips on the processor
 - Retains content when a router is powered down or restarted
 - Can store multiple versions of IOS software



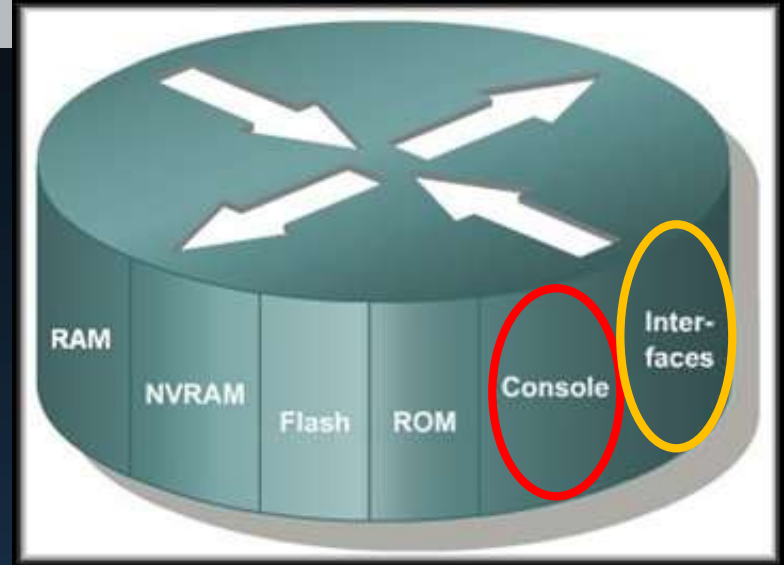
Router Components

- Read Only Memory (ROM):
 - Maintains instructions for power-on self test (POST) diagnostics
 - Stores the bootstrap program and the basic operating system software

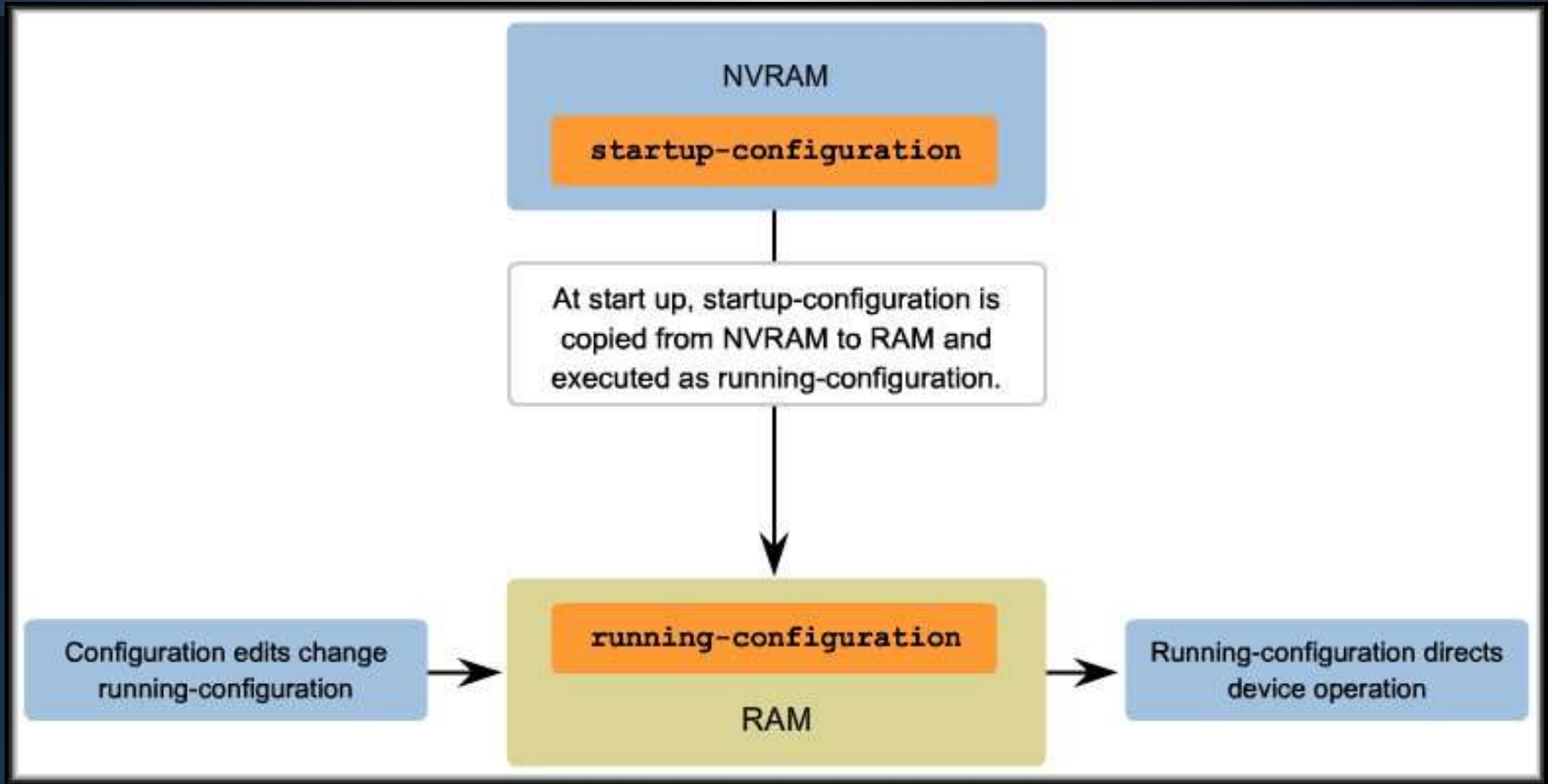


Router Components

- **Console:**
 - The console port provides physical access for the initial configuration of the router or configuration changes.
- **Interfaces:**
 - Connect routers to a network for packet entry and exit
 - They can be on the motherboard or on a separate module



Configuration Files



- Network devices depend upon two types of software:
 - Operating system (IOS)
 - Configuration file

Cisco IOS Modes

```
Router>ping 192.168.10.5
```

User EXEC

```
Router#show running-config
```

Privileged EXEC

```
Router(config)#Interface FastEthernet 0/0
```

Global Configuration

```
Router(config-if)#ip address 192.168.10.1 255.255.255.0
```

Interface Configuration

The prompt changes to denote the current CLI mode.

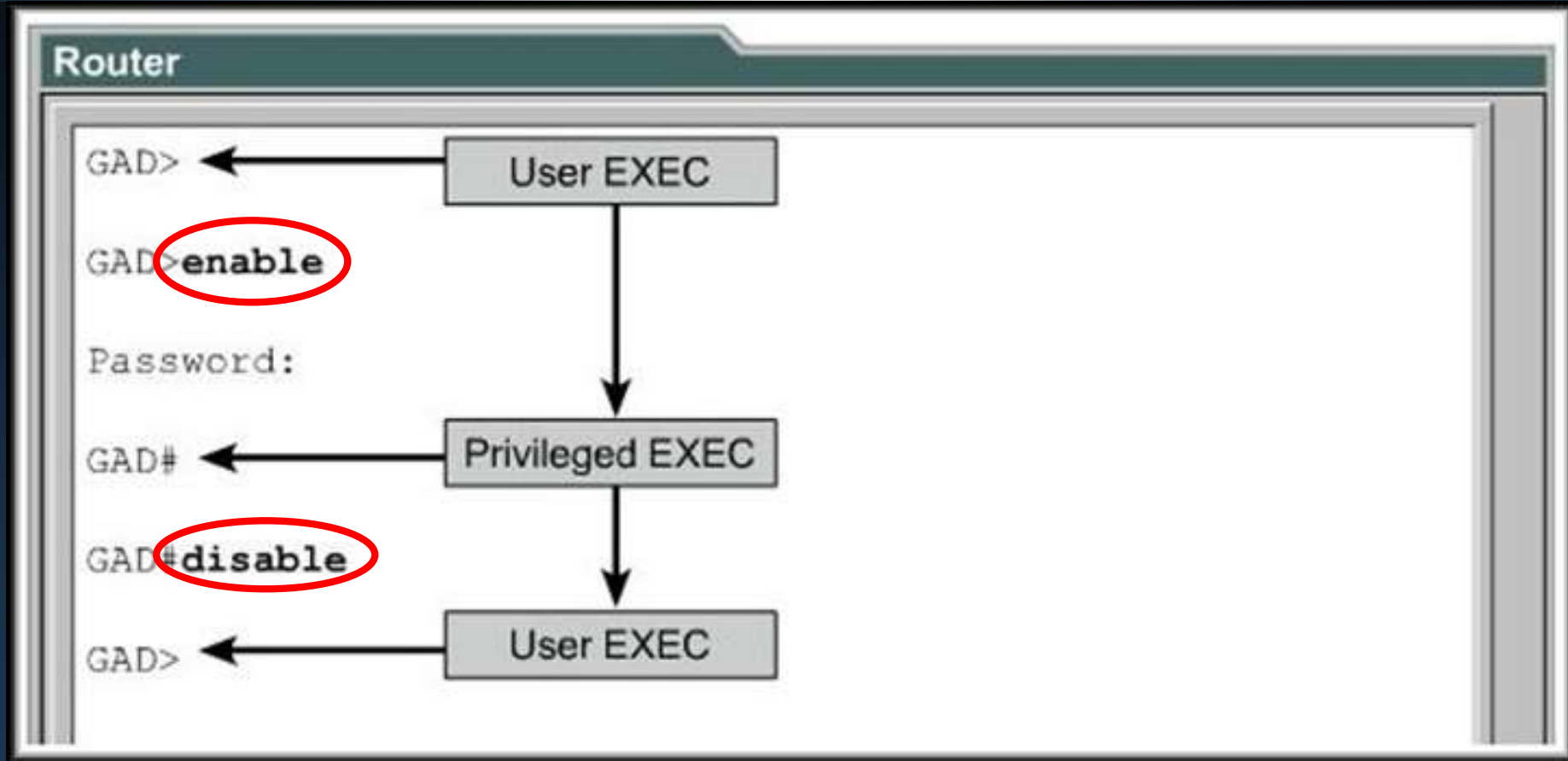
```
Switch>ping 192.168.10.9
```

```
Switch#show running-config
```

```
Switch(config)#Interface FastEthernet 0/1
```

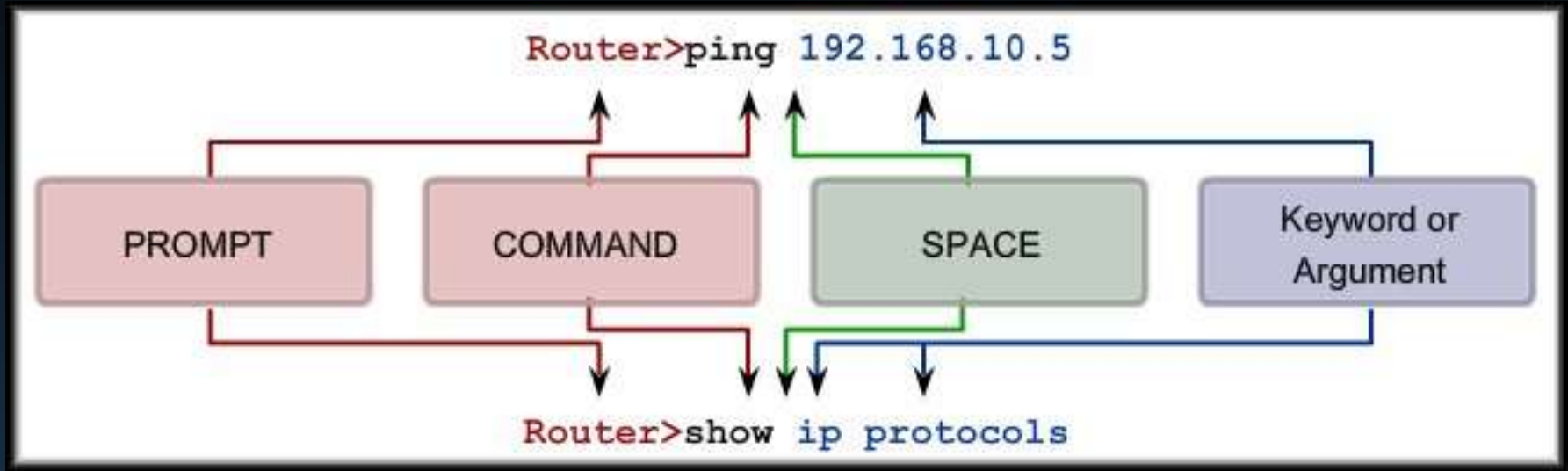
```
Switch(config-if)#Description connection to WEST LAN4
```

Moving Between User and Privileged EXEC



- Use *enable* to move from User to Privileged EXEC.
- Use *disable* or *end* to move back.

Basic IOS Command Structure



Router(config)#hostname MyRouter

MyRouter(config)#

Router(config)#interface fastethernet 0/0

Router(config-if)#

Router(config-if)# ip address 192.168.100.1 255.255.255.0

Router(config-if)#

Using CLI Help

Context Sensitive Help

Example of a sequence of commands using the CLI context sensitive help

```
Cisco#cl?  
clear clock  
Cisco#clock ?  
    set Set the time and date  
Cisco#clock set  
% Incomplete command.  
Cisco#clock set ?  
    hh:mm:ss Current Time  
Cisco#clock set 19:50:00  
% Incomplete command.
```

Command explanations

Incomplete command messages

Invalid input messages

Variable formats

```
Cisco#clock set 19:50:00 ?  
    <1-31> Day of the month  
    MONTH Month of the year  
Cisco#clock set 19:50:00 25 6  
                                     ^  
Invalid input detected at '^' marker.  
Cisco#clock set 19:50:00 25 June  
% Incomplete command.  
Cisco#clock set 19:50:00 25 June ?  
    <1993-2035> Year  
Cisco#clock set 19:50:00 25 June 2007  
Cisco#
```

Using CLI Help

Command Syntax Check Help

The IOS returns a help message indicating that required keywords or arguments were left off the end of the command:

```
Switch#>clock set
% Incomplete command.
Switch#clock set 19:50:00
% Incomplete command.
```

The IOS returns a help message to indicate that there were not enough characters entered for the command interpreter to recognize the command.

```
Switch#c
% Ambiguous command: 'c'
```

The IOS returns a "^" to indicate where the command interpreter can not decipher the command:

```
Switch#clock set 19:50:00 25 6
                        ^
% Invalid input detected at '^' marker.
```

Using CLI Help

- Hot Keys and Shortcuts:
 - CLI Line Editing:
 - Several outlined in the text.
 - **Backspace** most often used.
 - Note that the **Delete** key is **NOT** recognized to delete invalid characters.

Using CLI Help

- Hot Keys and Shortcuts:
 - At the ----More---- Prompt:
 - If the screen fills up with a display, the More prompt will be displayed.

At the "-----More-----" prompt

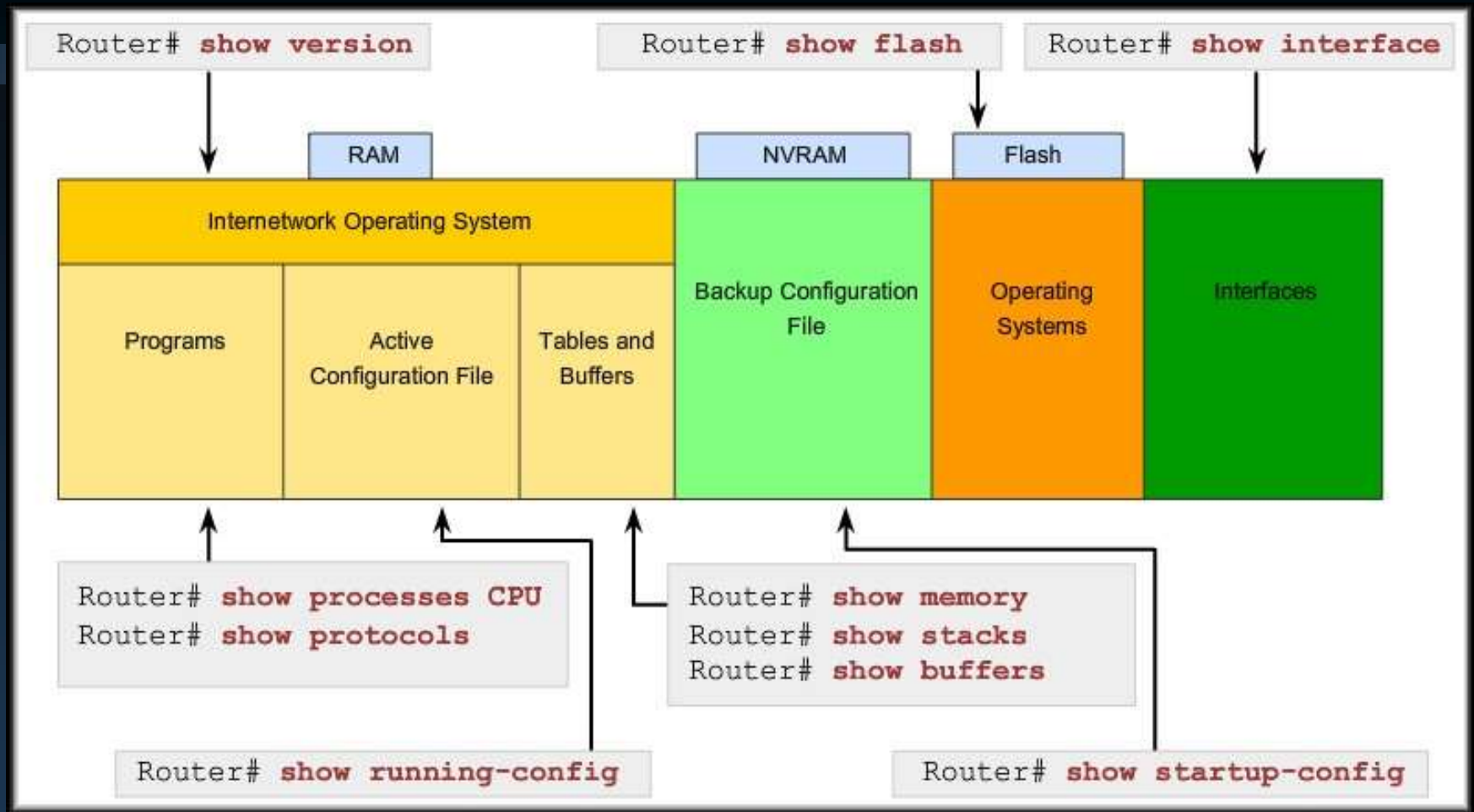
The Enter Key	Displays the next line.
Space Bar	Displays the next screen.
Any other alphanumeric key	Returns to the EXEC prompt.

Using CLI Help

- Break Keys:
 - If you wish to abort a process or command.....

Break Keys	
Ctrl-C	When in any configuration mode, ends the configuration mode and returns to privileged EXEC mode. When in setup mode, aborts back to the command prompt.
Ctrl-Z	When in any configuration mode, ends the configuration mode and returns to privileged EXEC mode.
Ctrl-Shift-6	All-purpose break sequence. Use to abort DNS lookups, traceroutes, pings.

IOS Examination Commands



- In order to verify and troubleshoot network operation, we must examine the operation of the devices.
- The basic examination command is the **show** command.

IOS Examination Commands – show version

```
Router
Cisco#show version

Cisco Internetwork Operating System Software
IOS (tm) C2600 Software (C2600-JK8S-M), Version
12.2(12c), RELEASE SOFTWARE (fc1)
Copyright (c) 1986-2003 by cisco Systems, Inc.
Compiled Wed 05-Feb-03 16:36 by kellythw
Image text-base: 0x8000808C, data-base: 0x8156F2AC

ROM: System Bootstrap, Version 11.3(2)XA4, RELEASE
SOFTWARE (fc1)
R2 uptime is 4 weeks, 2 days, 17 hours, 9 minutes
System returned to ROM by reload
System image file is "flash:c2600-jk8s-mz.122-12c.bin"

cisco 2620 (MPC860) processor (revision 0x102) with
59392K/6144K bytes of memory

----- output omitted -----

Basic Rate ISDN software, Version 1.1.
1 FastEthernet/IEEE 802.3 interface(s)
2 Low-speed serial(sync/async) network interface(s)
1 ISDN Basic Rate interface(s)
32K bytes of non-volatile configuration memory.

16384K bytes of processor board System flash
(Read/Write)

Configuration register is 0x2102
```

IOS Version,
Name

RAM

INTERFACES

FLASH

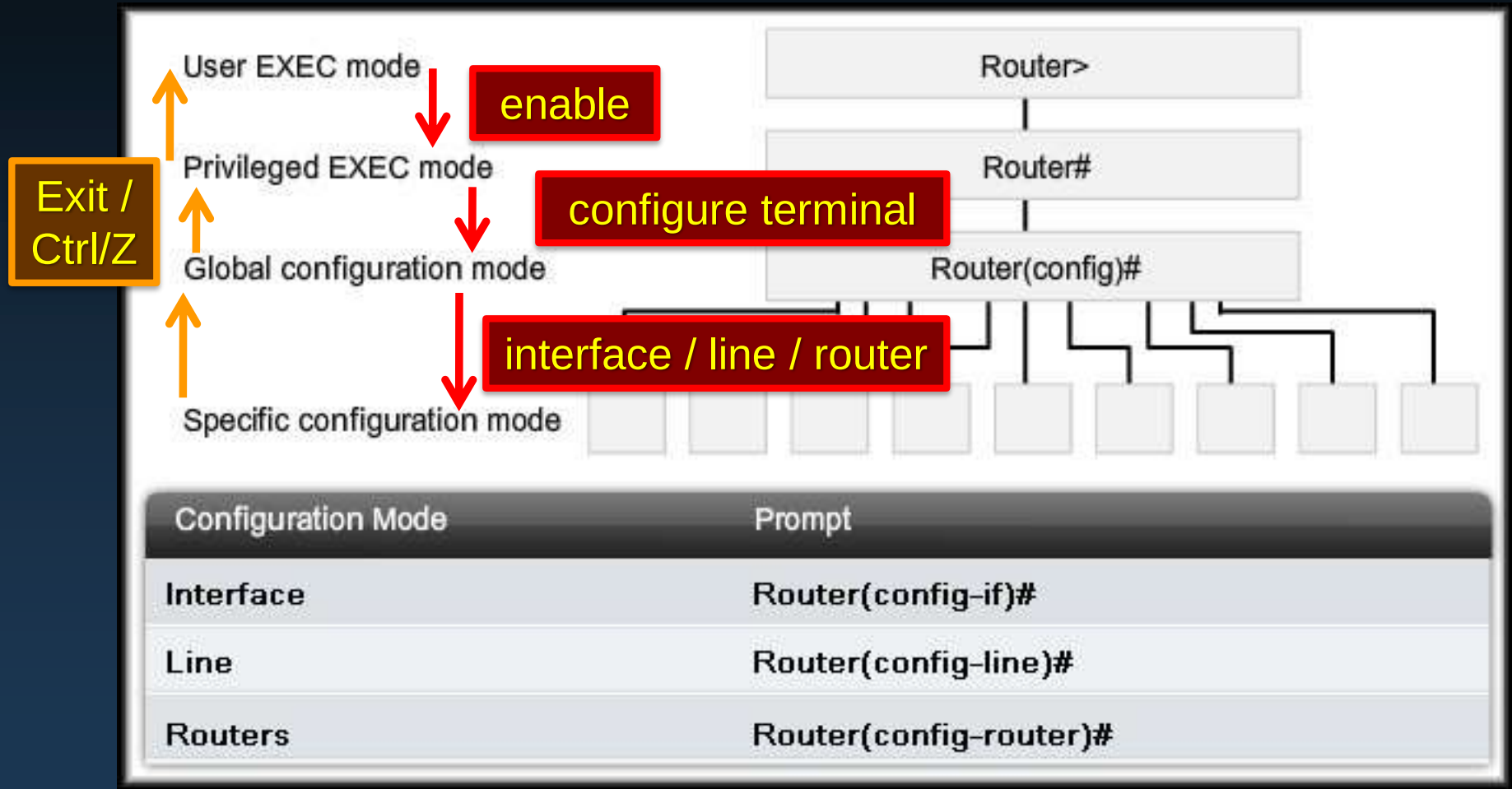
Configuration
Register

IOS Examination Commands – show flash

```
Router
BHM#show flash
PCMCIA flash directory:
File Length Name/status
1 6007232 c1700-bnsy-1.212-11.p
[6007296 bytes used, 284160 available, 6291456
total]
6144K bytes of processor board PCMCIA flash (Read
ONLY)
BHM#
```

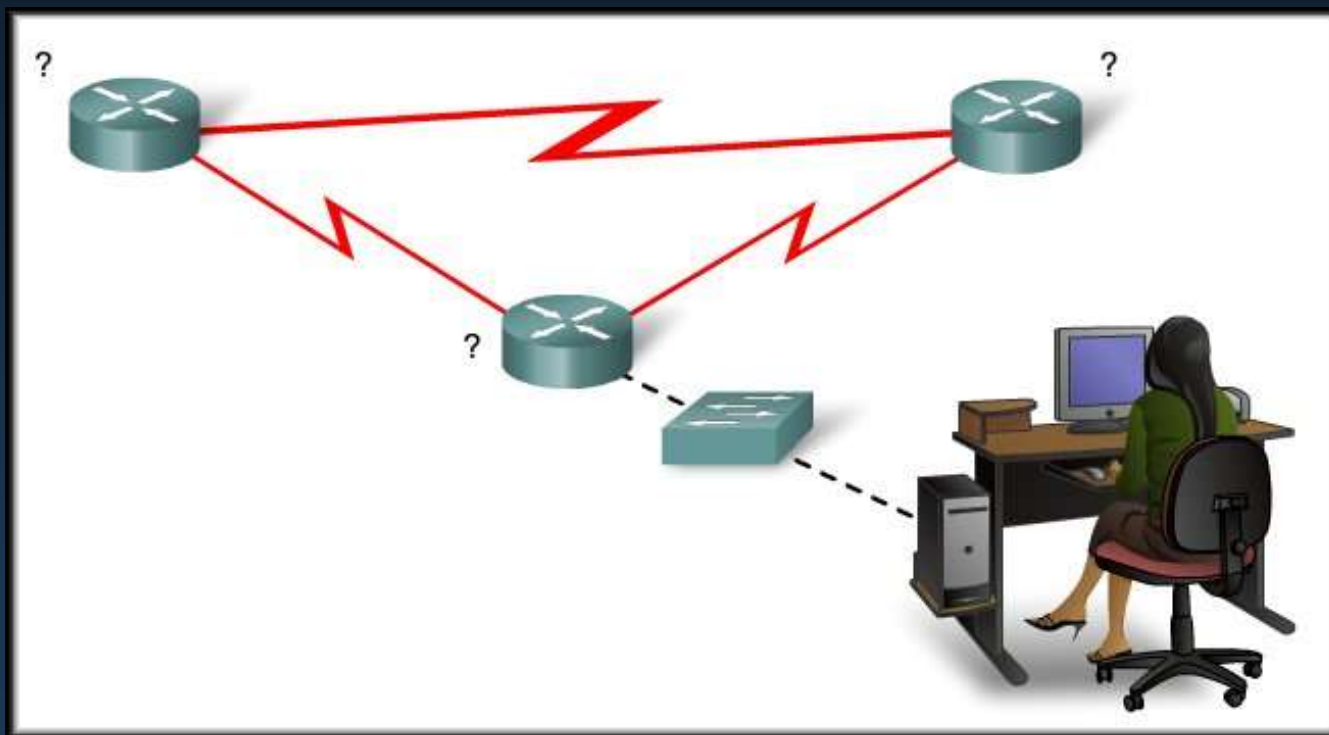
- Use the *show flash* command to verify that the router has sufficient memory to load a new Cisco IOS software image.

IOS Configuration Modes



Configuring and Testing Your Network

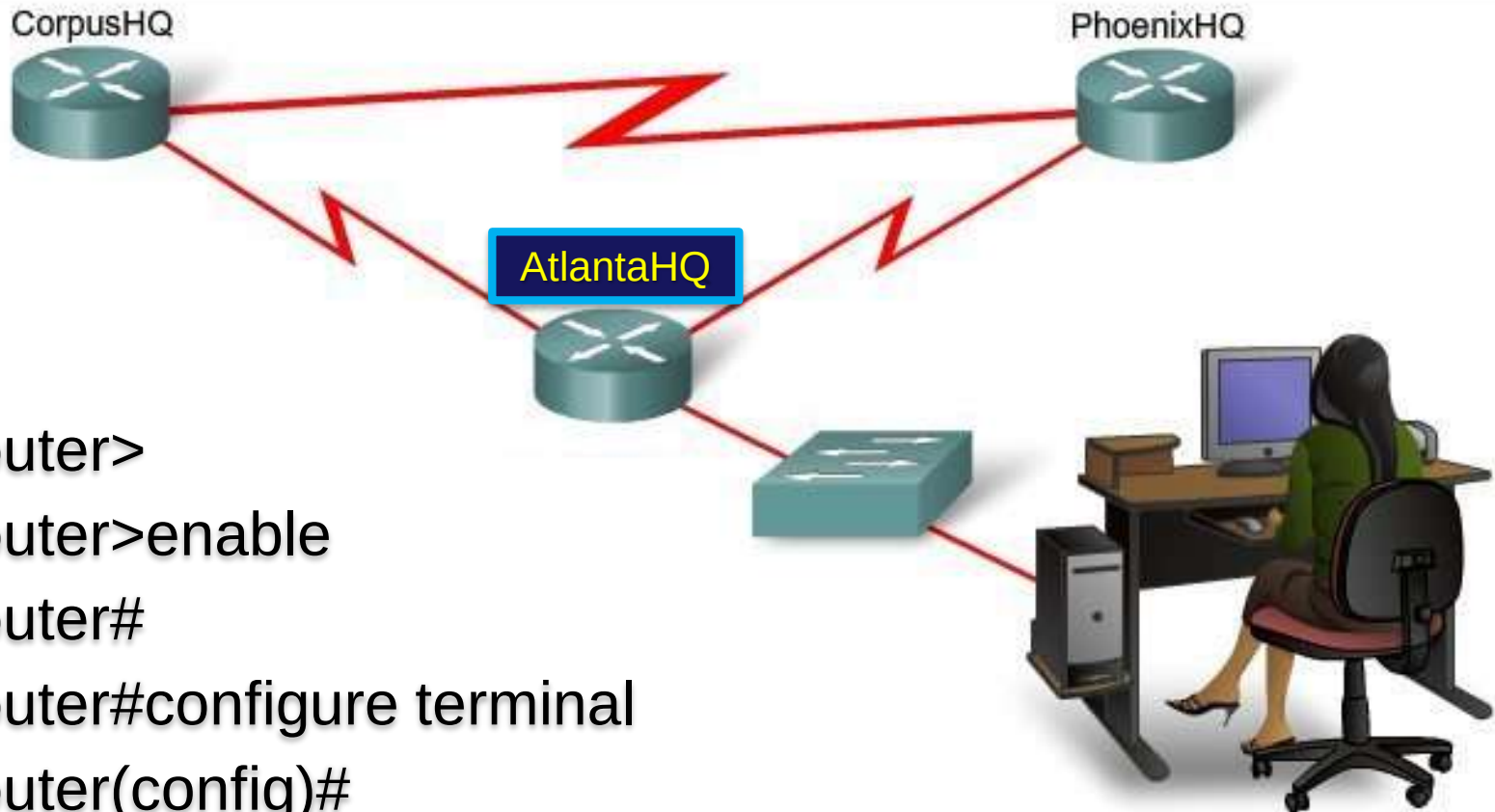
Applying a Basic Configuration



Naming Devices

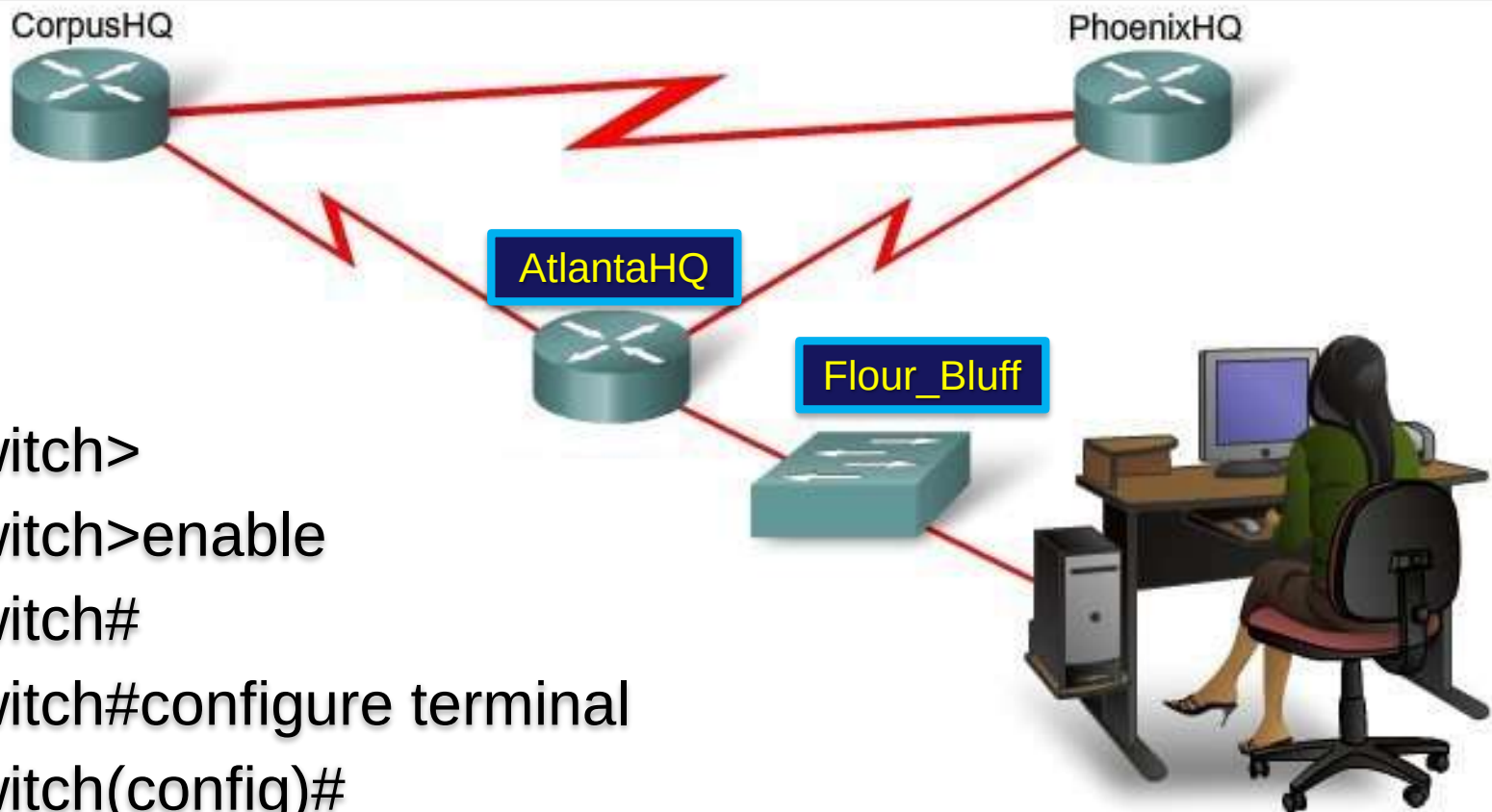
- Importance of a Device Name:
 - Default “Router” or “Switch”
 - Host Name appears as the CLI prompt.
 - Confirmation that the correct device has been accessed.
 - Naming convention makes maintenance easier.
 - Start with a letter
 - No spaces
 - End with letter or digit
 - Only letters, digits and dashes
 - Maximum of 63 characters

Naming Devices



```
Router>  
Router>enable  
Router#  
Router#configure terminal  
Router(config)#  
Router(config)#hostname AtlantaHQ  
AtlantaHQ(config)#
```

Naming Devices



```
Switch>  
Switch>enable  
Switch#  
Switch#configure terminal  
Switch(config)#  
Switch(config)#hostname Flour_Bluff  
Flour_Bluff(config)#
```

Limiting Device Access

- **Configuring Passwords:**

- Passwords are the primary defense against unauthorized access to network devices.

Every device should have locally configured passwords to limit access.

- **Console password:**

- Limits device access using the console connection

- **Enable password:**

- Limits access to the privileged EXEC mode

- **Enable secret password:**

- Encrypted, limits access to the privileged EXEC mode

- **VTY password:** Limits device access using Telnet / SSH

Limiting Device Access

Console Password

```
Router(config)#line console 0  
Router(config-line)#login  
Router(config-line)#password cisco
```



Virtual Terminal Password

```
Router(config)#line vty 0 4  
Router(config-line)#login  
Router(config-line)#password cisco
```



Enable Password

```
Router(config)#enable password san-fran
```



Enable Password

```
Router(config)#enable secret <password>
```



Limiting Device Access

- Encrypting Password Display:
 - The *service password-encryption* command prevents ALL passwords from showing up as plain text when viewing the configuration files.

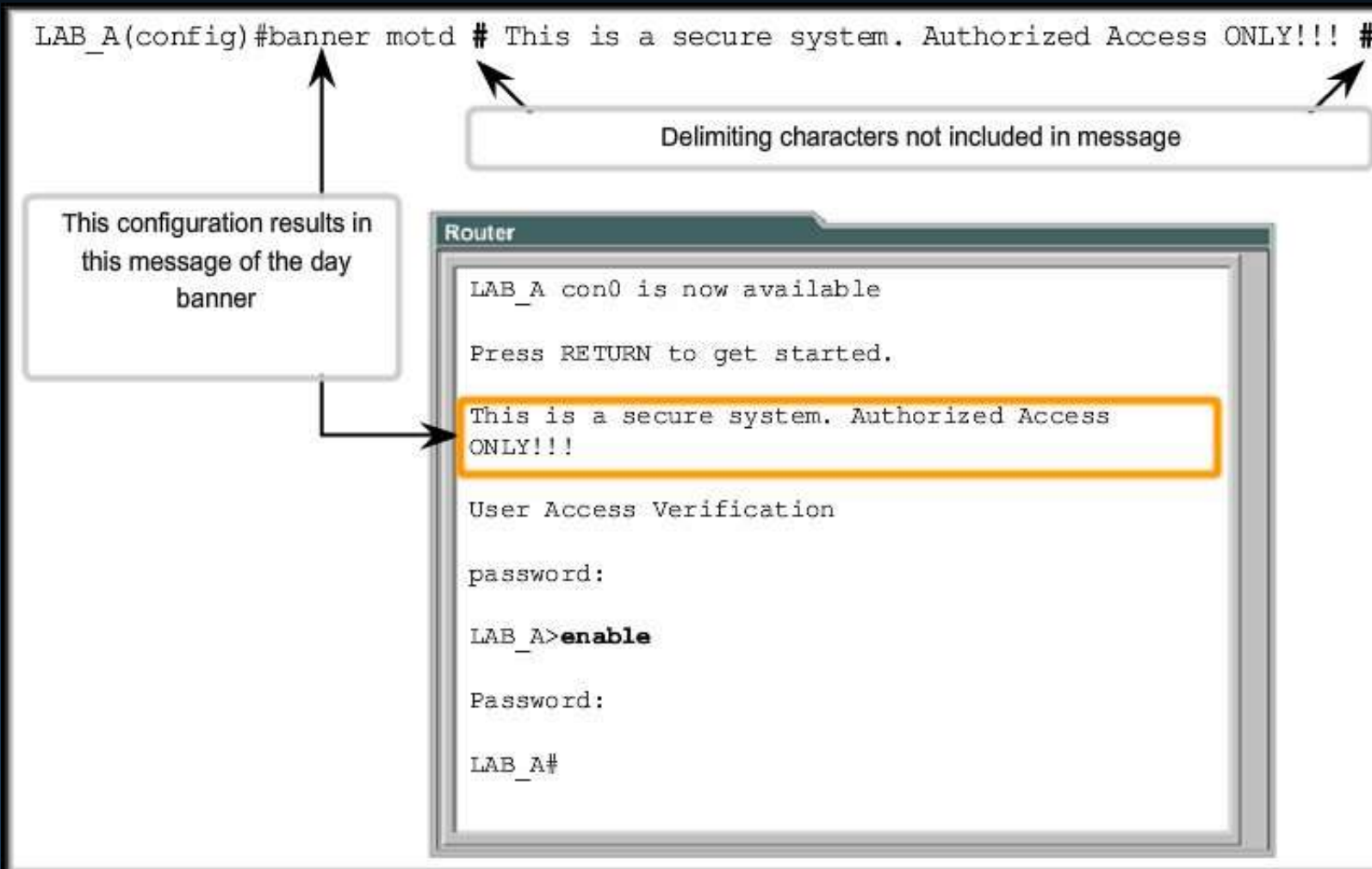
Perform Password Encryption

```
Router(config)#service password-encryption  
(set passwords here)  
Router(config)#no service password-encryption
```

service password-encryption uses a Cisco Level 7 encryption which is very easy to decrypt.

Limiting Device Access

- Login Banner:



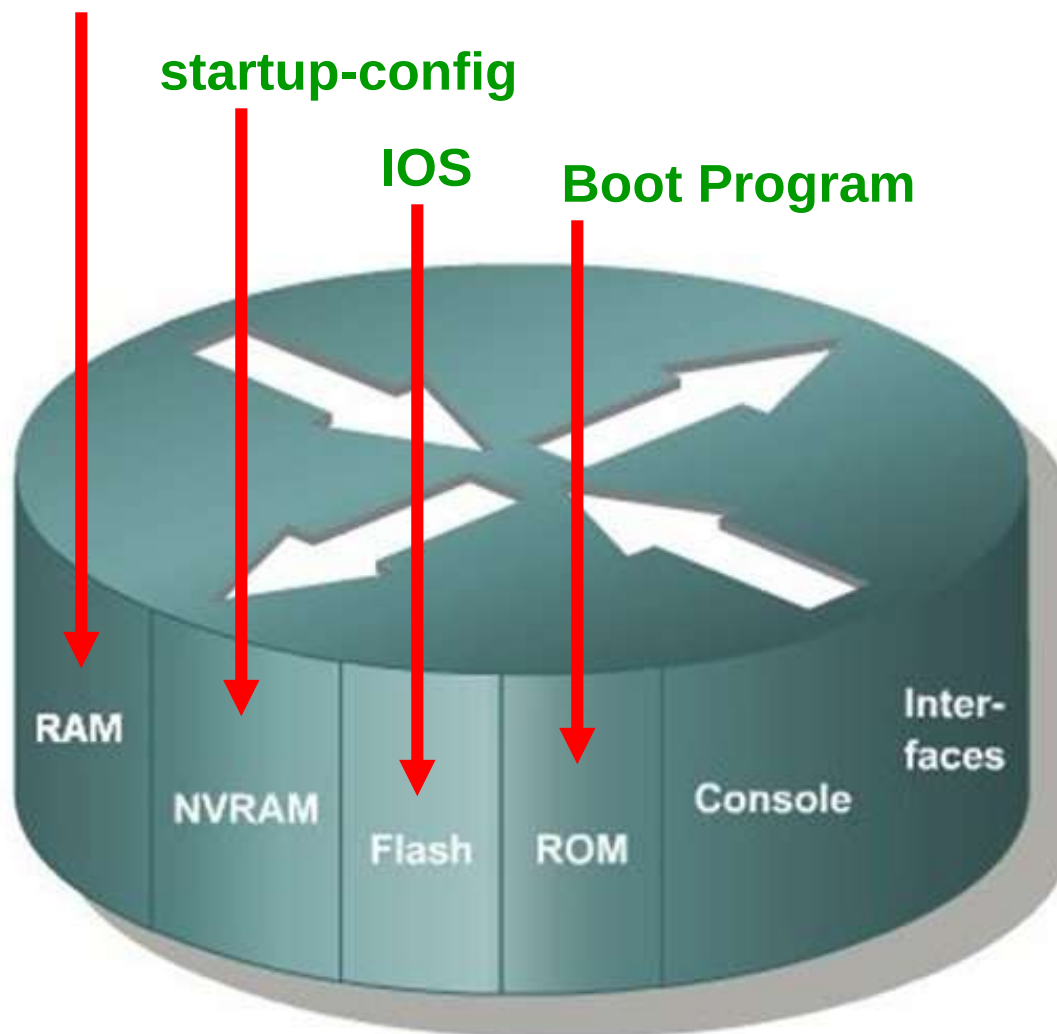
Managing Configuration Files

running-config
(IOS running)

startup-config

IOS

Boot Program



Managing Configuration Files

```
Router#show startup-config  
%% Non-volatile configuration memory is not present
```

```
Router#show running-config  
Building configuration...
```

```
Current configuration:  
!  
version 12.0  
service timestamps debug uptime  
service timestamps log uptime  
no service password-encryption  
!  
hostname Router  
!  
ip subnet-zero  
!  
!  
interface Ethernet0  
  no ip address  
  no ip directed-broadcast  
  shutdown  
!  
interface Serial0  
  no ip address  
  no ip directed-broadcast  
  shutdown
```

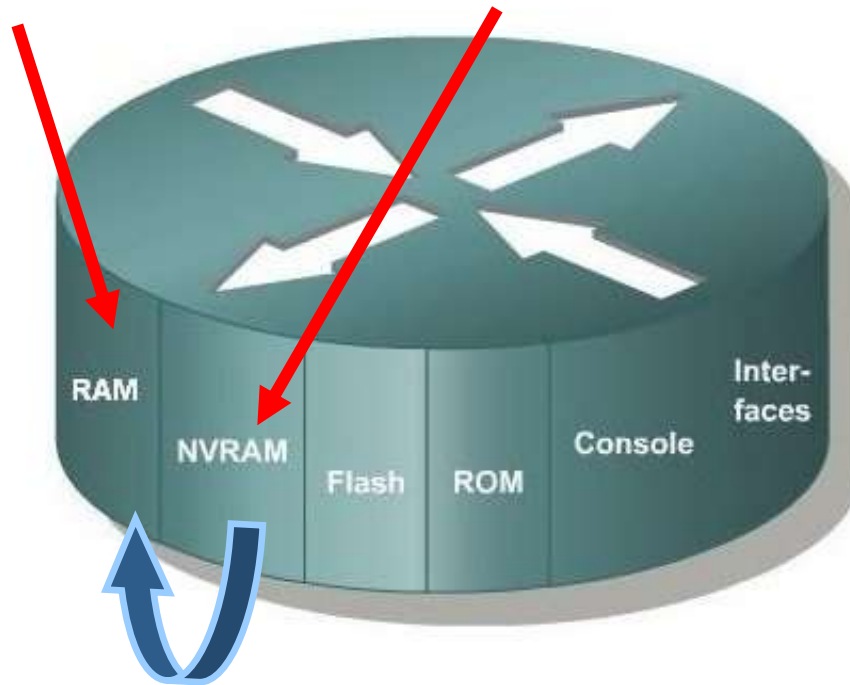
No startup-config
file in NVRAM

Default running-config
file, created in RAM

Managing Configuration Files

running-config
resides here

startup-config
resides here

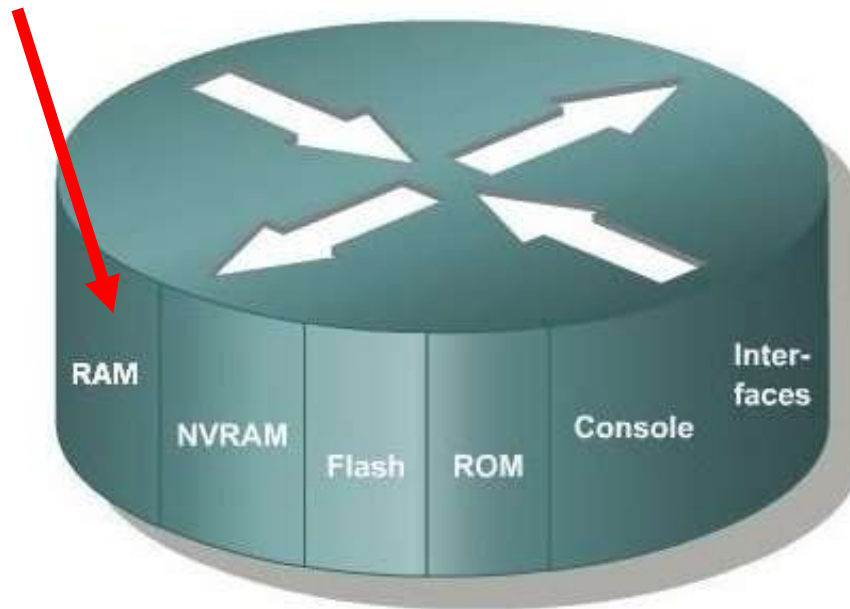


At boot up, running-config is *replaced* by startup-config....

Managing Configuration Files

running-config
resides here

If the router loses power or
reboots, everything in RAM is lost
including the running-config file.

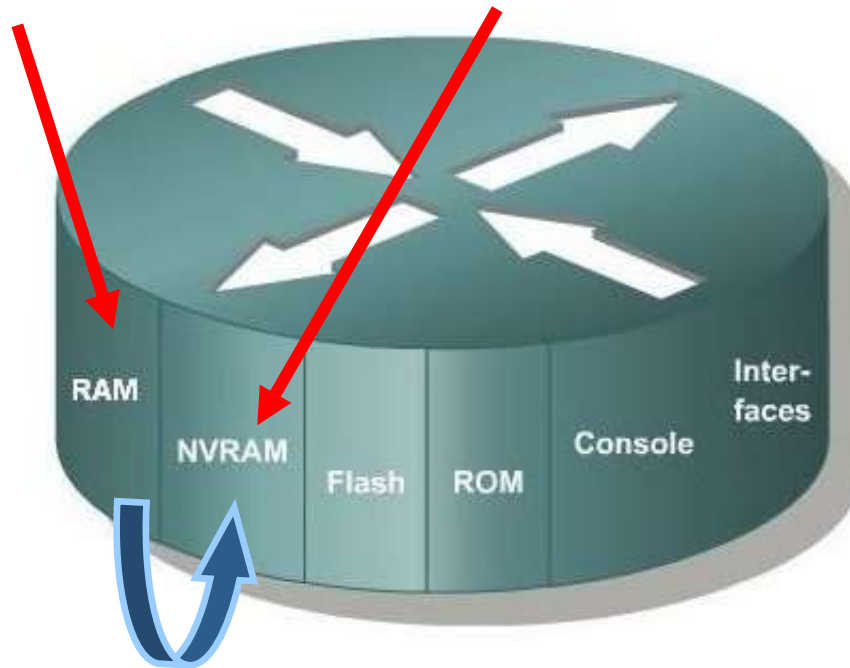


Changes to the router are automatically put in
the running-config file.

Managing Configuration Files

running-config
resides here

startup-config
resides here



To make sure that router changes are saved...
copy running-config startup-config

Managing Configuration Files

```
Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
```

```
Router#show startup-config
```

```
!
version 12.0
service timestamps debug uptime
service timestamps log uptime
no service password-encryption
!
hostname Router
!
!
ip subnet-zero
!
interface Ethernet0
  no ip address
  no ip directed-broadcast
  shutdown
!
```

The startup-config file is now identical to running-config and the router will have these changes if the router reboots.

Managing Configuration Files

```
Router# copy running-config startup-config
```

OR

```
Router# copy running startup
```

OR

```
Router# copy run start
```

*Any usage of the command or parameters,
so that they are
still uniquely recognizable.*

Managing Configuration Files

- Returning the Device to its Original Configuration:
 - **reload** command
 - Works **ONLY IF** the running configuration has **NOT** been copied to the startup configuration.

```
Router# reload
```

```
System configuration has been modified. Save?  
[yes/no]: n
```

```
Proceed with reload? [confirm] Press Enter
```

```
*Apr 13 01:34:15.758: %SYS-5-RELOAD: Reload  
requested by console.
```

```
Reload Reason: Reload Command.
```

Managing Configuration Files

- Backing Up Configurations Offline:
 - TFTP server
 - CD
 - USB Memory Stick
 - Text Files using the Clipboard.
 - Text files using HyperTerminal capture.
- Restoring Configurations:
 - TFTP Server
 - Paste to Host from a text file.

Managing Configuration Files

- A **TFTP server** will allow image and configuration uploads and downloads over the network.
- The TFTP server can be another router, or it can be a host system.

Managing Configuration Files

Flash

RAM

TFTP Server

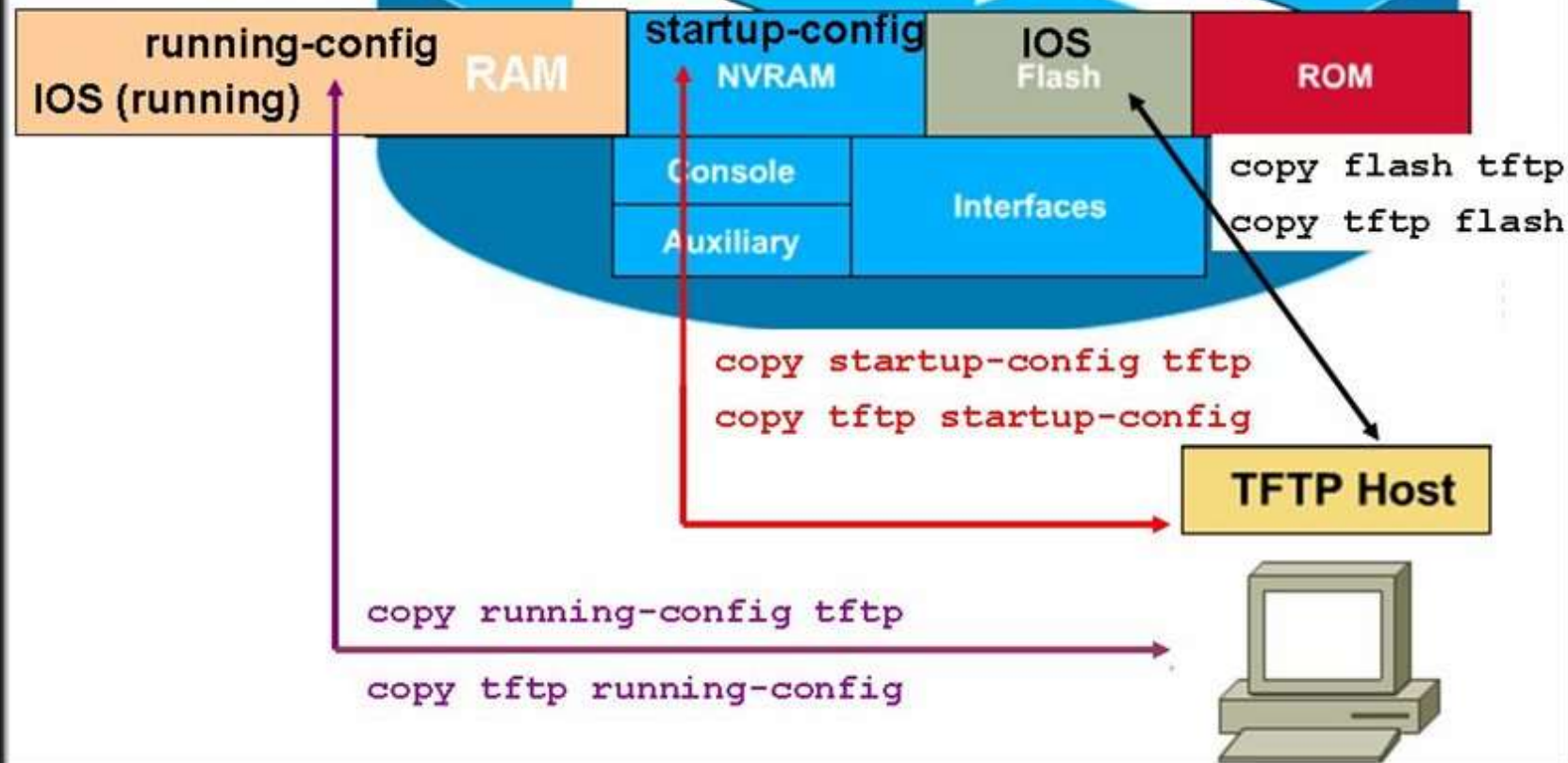
Command

```
Router#ping tftp - address
Type escape sequence to abort
Sending 5, 100-byte ICMP Echoes to 210.93.105.1
timeout is 2 seconds:
!!!!.!
Success rate is 80 percent (4/5)
round trip min/avg/max = 68/68/168 ms
```

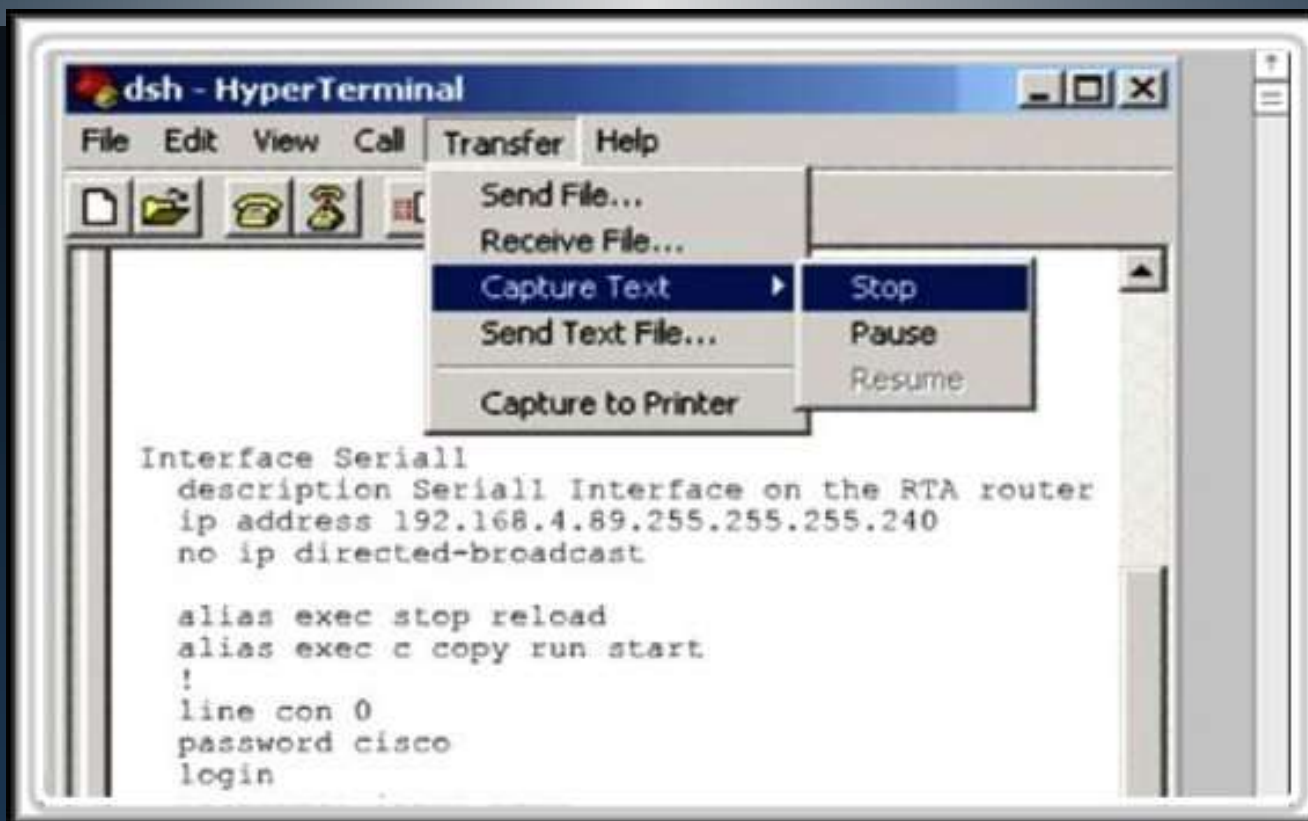
- Once it is installed or before you attempt to use it, **ALWAYS** make sure that the connectivity is there....

Managing Configuration Files

What can be copied?



Managing Configuration Files



- Configuration files can be captured and saved **in text format** using the text capture function of HyperTerminal (or any other emulator that has the function).

Configuring Interfaces

All interfaces are accessed by issuing the `interface` command at the global configuration prompt.

In the following commands, the `type` argument includes serial, ethernet, fastethernet, and others:

```
Router(config)#interface type port
Router(config)#interface type slot/port
Router(config)#interface type slot/subslot/port
```

**Router
Interfaces**

The following command is used to administratively turn off the interface:

```
Router(config-if)#shutdown
```

The following command is used to turn on an interface that has been shutdown:

```
Router(config-if)#no shutdown
```

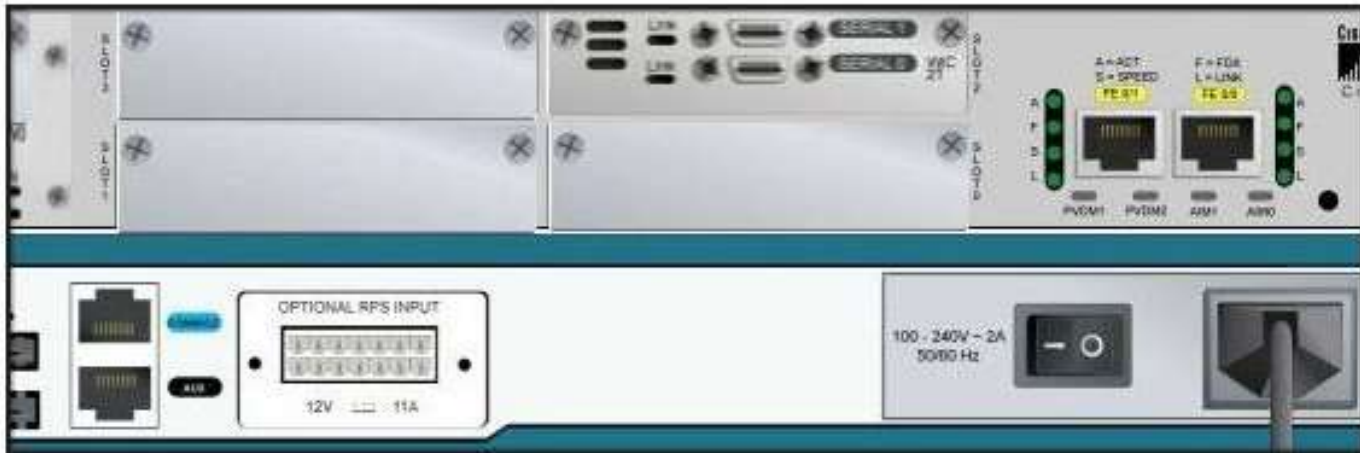
The following command is used to quit the current interface configuration mode:

```
Router(config-if)#exit
```

When the configuration is complete, the interface is enabled and interface configuration mode is exited.

Configuring Interfaces

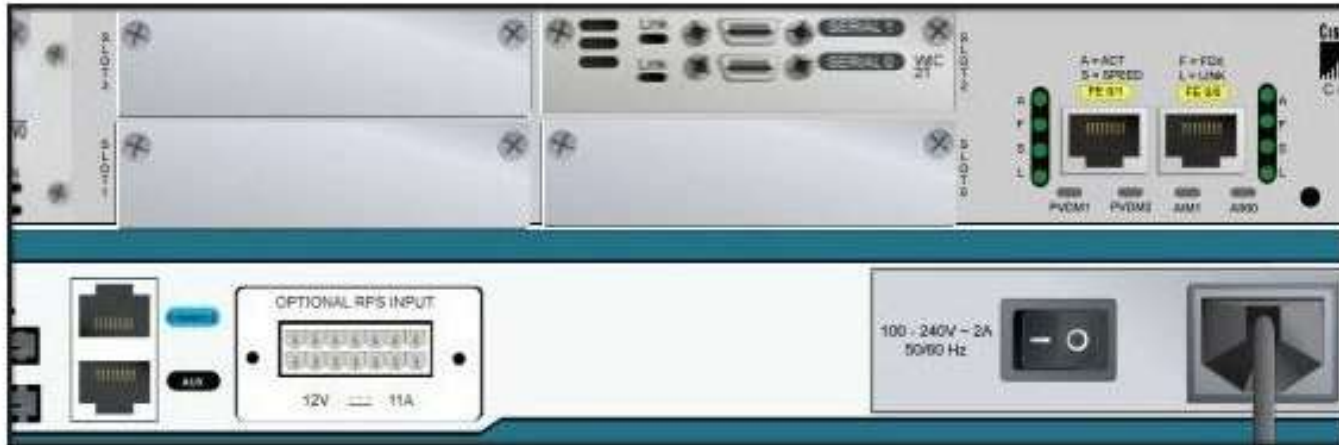
- Configuring a Router Ethernet Interface:



```
Router#configure terminal
Router(config)#
Router(config)#interface FastEthernet 0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
```

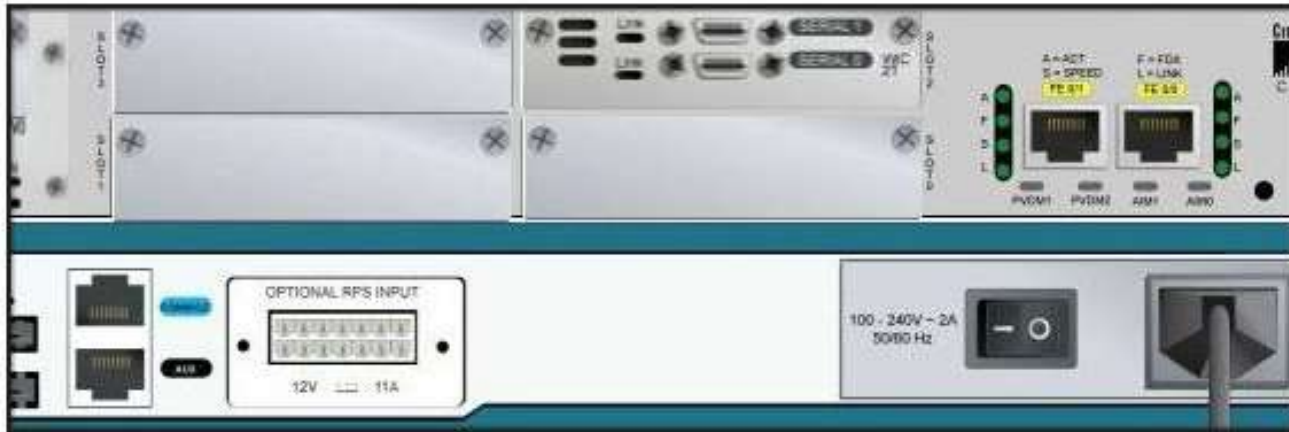
Configuring Interfaces

- Configuring a Router Serial Interface:



```
Router#configure terminal
Router(config)#
Router(config)#interface Serial 0/2/0
Router(config-if)#ip address 192.168.11.1 255.255.255.0
Router(config-if)#clock rate 56000
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
```

Describing Router Interfaces



```
Router(config)#interface FastEthernet 0/0
Router(config-if)#description Building B Sales LAN
Router(config-if)#exit
```

Description is all text after this space

Interface description used for internal network documentation

```
Router(config)#interface Serial 0/0/0
Router(config-if)#description To Perth CKT-PT27834365-01
Router(config-if)#exit
```

Description is all text after this space

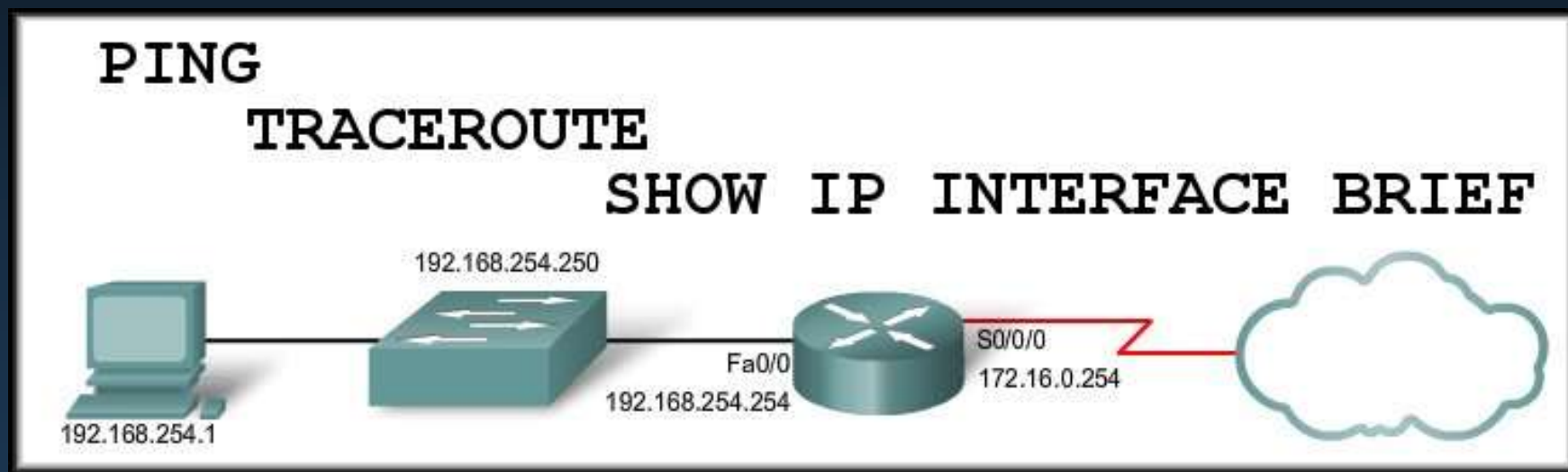
Configuring Interfaces

- Configuring a Switch:

```
Switch#configure terminal
Switch(config)#
Switch(config)#hostname Flour_Bluff
Flour_Bluff(config)#
Flour_Bluff(config)#interface FastEthernet 0/1
Flour_Bluff(config-if)#description To TAM switch
Flour_Bluff(config-if)#interface vlan 1
Flour_Bluff(config-if)#ip address 192.168.1.2 255.255.255.0
Flour_Bluff(config-if)#no shutdown
Flour_Bluff(config-if)#exit
Flour_Bluff(config)#ip default-gateway 192.168.1.1
Flour_Bluff(config)#exit
Flour_Bluff#
```

Configuring and Testing Your Network

Verifying Connectivity



Test the Stack

- PING:
 - From the command window on a PC.



```
C:\WINDOWS\system32\cmd.exe

C:\>ping 192.168.100.1

Pinging 192.168.100.1 with 32 bytes of data:

Reply from 192.168.100.1: bytes=32 time=1ms TTL=63
Reply from 192.168.100.1: bytes=32 time=1ms TTL=63
Reply from 192.168.100.1: bytes=32 time=1ms TTL=63
Reply from 192.168.100.1: bytes=32 time=1ms TTL=63

Ping statistics for 192.168.100.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>
```


Test the Stack

- PING:
 - From the router IOS.

```
R2-Central>ping 172.16.1.1
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 172.16.1.1, timeout is 2 seconds:
```

```
!!!!!
```

```
R2-Central>ping 172.16.1.20
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 172.16.1.20, timeout is 2 seconds:
```

```
.....
```

```
R2-Central>ping 202.163.24.5
```

```
Type escape sequence to abort.
```

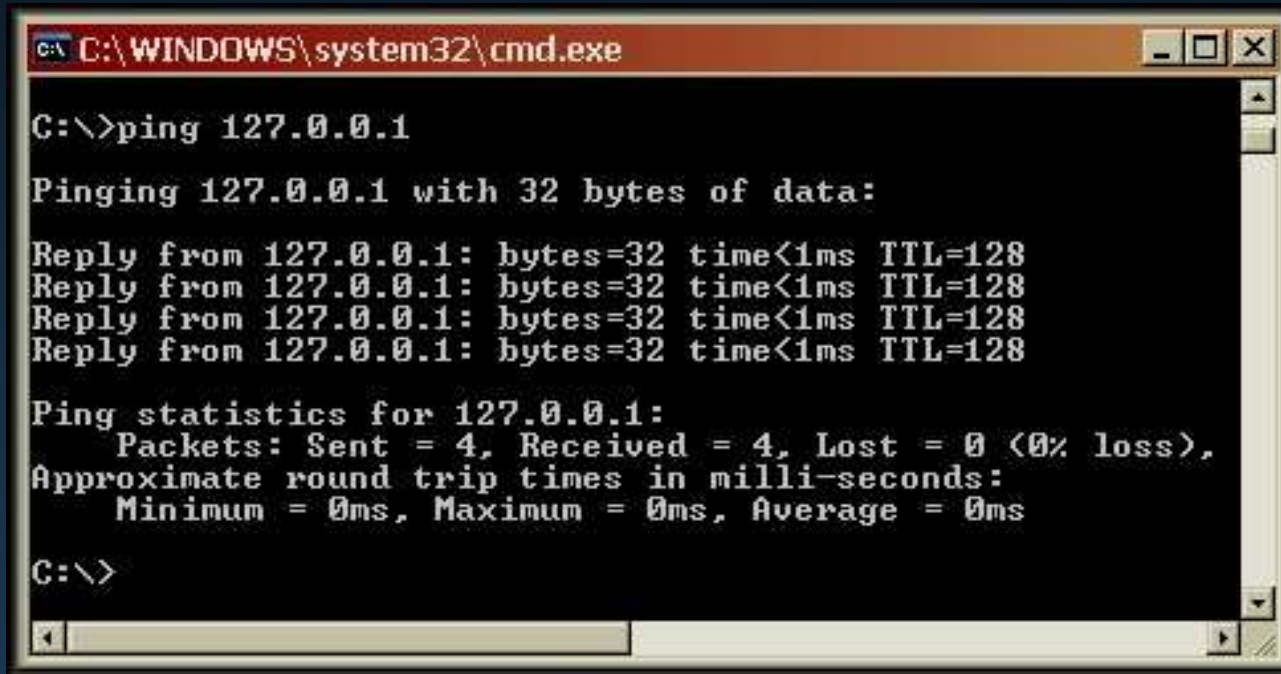
```
Sending 5, 100-byte ICMP Echos to 202.163.24.5, timeout is 2 seconds:
```

```
U.U.U
```

```
Success rate is 0 percent (0/5)
```

Test the Stack

- PING:
 - Testing the Loopback (127.0.0.1).



```
C:\WINDOWS\system32\cmd.exe

C:\>ping 127.0.0.1

Pinging 127.0.0.1 with 32 bytes of data:

Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 127.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

If successful, it means that the protocol stack on the PC is functioning properly from the Network through the Physical Layers.

Testing the Local NIC Assignment

```
IP Address. . . . . : 10.0.0.5  
Subnet Mask . . . . . : 255.255.255.0  
Default Gateway. . . . : 10.0.0.254
```



Verify the host NIC address is bound and ready for transmitting signals across the media by pinging its own IP address.

Testing the Router Interfaces



```
Router1#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	192.168.254.254	YES	NVRAM	up	up
FastEthernet0/1	unassigned	YES	unset	down	down
Serial0/0/0	172.16.0.254	YES	NVRAM	up	up
Serial0/0/1	unassigned	YES	unset	administratively down	down

```
Router1#
```

```
Router1#ping 192.168.254.1
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 192.168.254.1, timeout is 2 seconds:
```

```
!!!!
```

```
Success rate is 100 percent (5/5) round-trip min/avg/max = 1/2/4 ms
```

```
Router1#
```

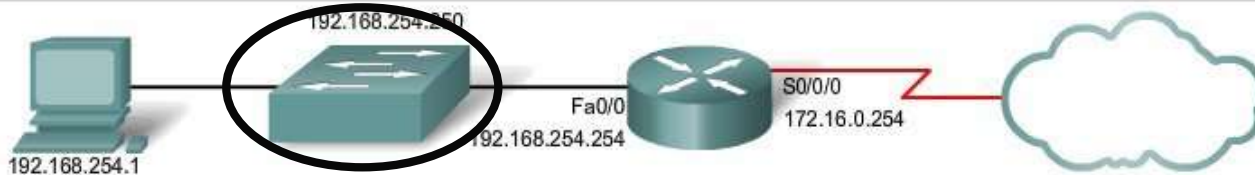
```
Router1#traceroute 192.168.0.1
```

```
Type escape sequence to abort.
```

```
Tracing the route to 192.168.0.1
```

1	172.16.0.253	8 msec	4msec	8 msec
2	10.0.0.254	16 msec	16 msec	8 msec
3	192.168.0.1	16 msec	*	20 msec

Testing the Switch Interfaces



```
Switch1#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/1	unassigned	YES	manual	up	up
.					
.					
FastEthernet0/24	unassigned	YES	manual	down	down
Vlan1	192.168.254.250	YES	manual	up	up

```
Switch1#
```

```
Switch1#ping 192.168.254.1
```

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.254.1, timeout is 2 seconds:
!!!!

Success rate is 100 percent (5/5) round-trip min/avg/max = 1/2/4 ms

```
Switch1#
```

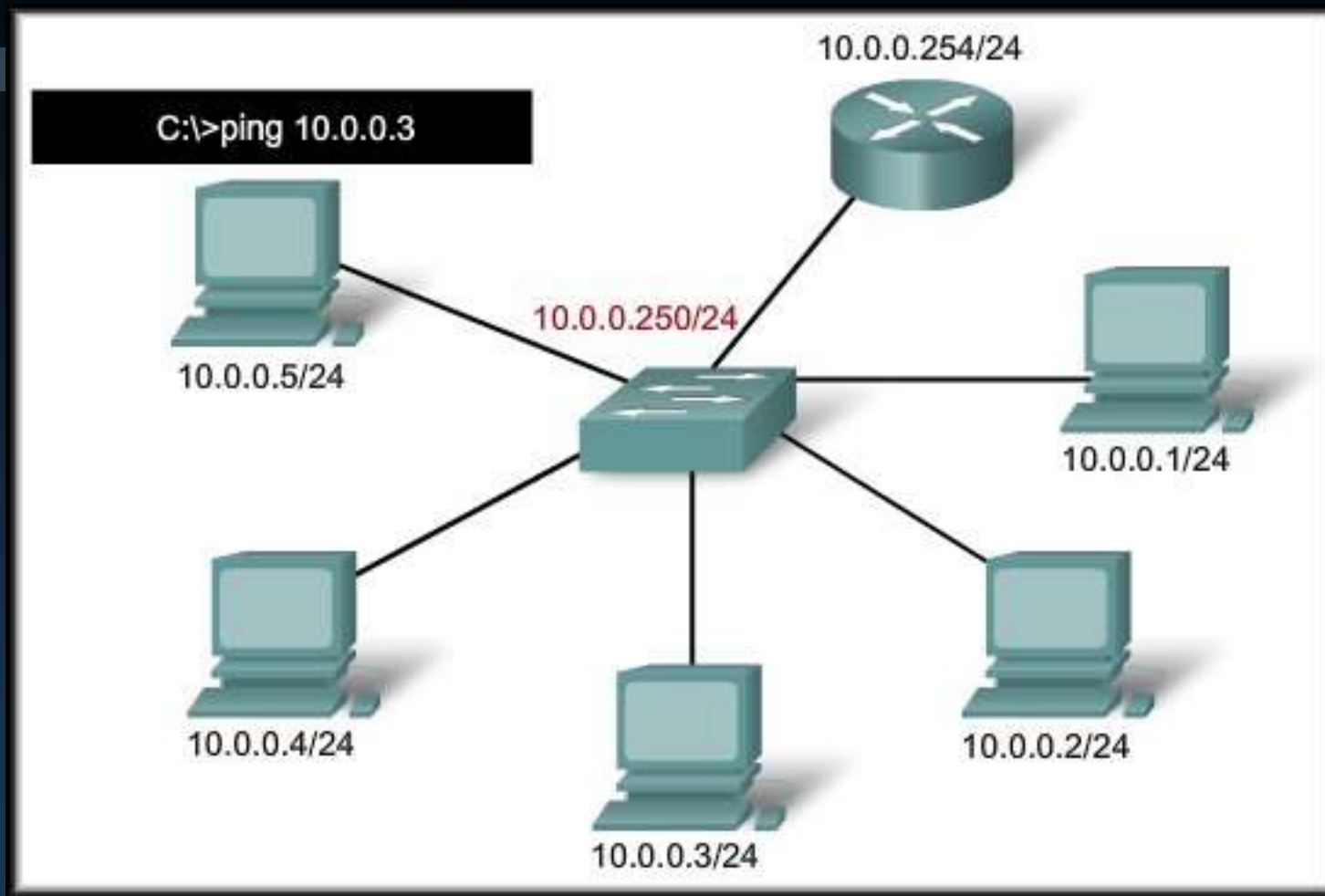
```
Switch1#traceroute 192.168.0.1
```

Type escape sequence to abort

Tracing the route to 192.168.0.1

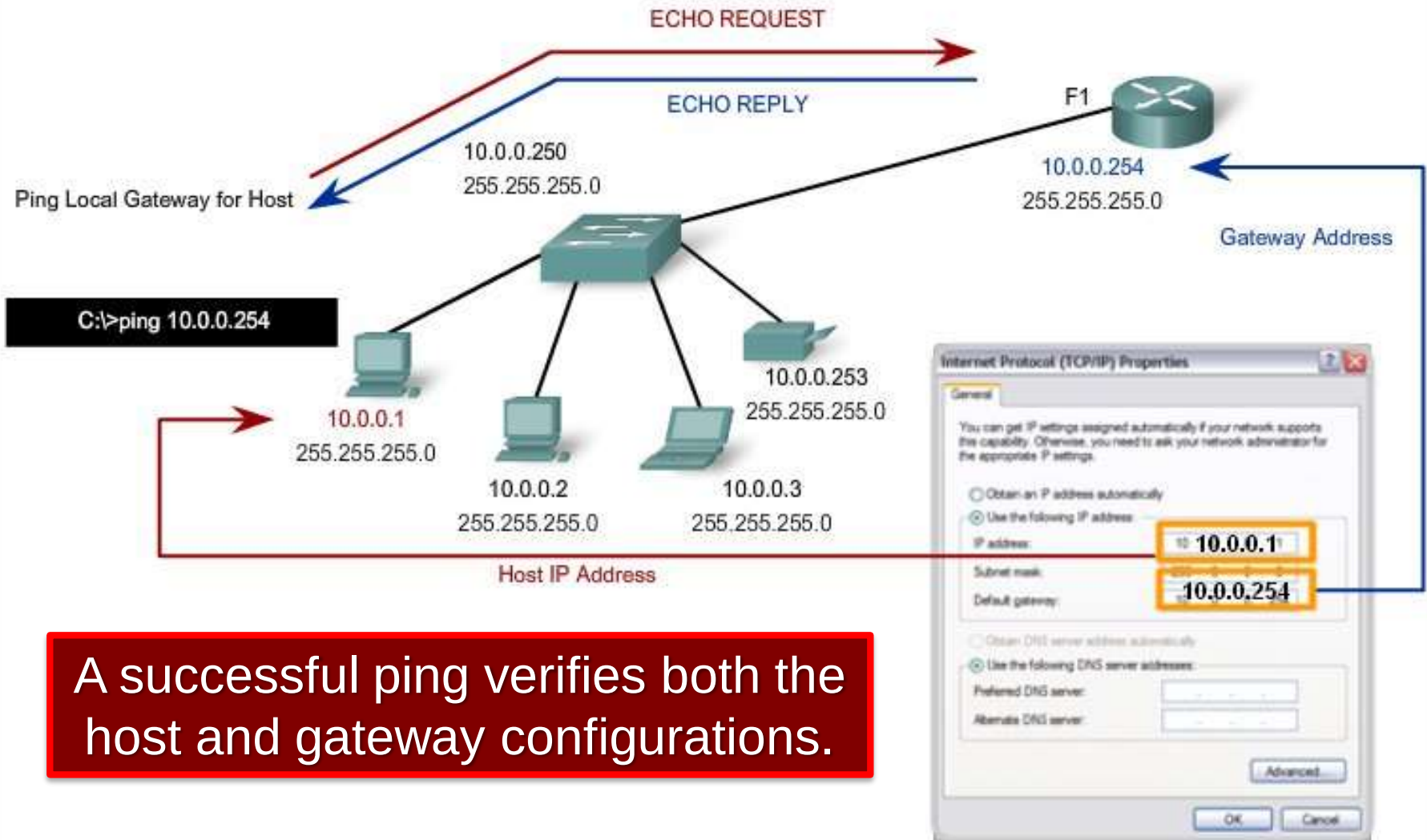
1	192.168.254.254	4 msec	2 msec	3 msec
2	172.16.0.253	8 msec	4 msec	8 msec
3	10.0.0.254	16 msec	16 msec	8 msec
4	192.168.0.1	16 msec	*	20 msec

Testing the Local Network



Successfully pinging another host on the same subnet verifies that both the local host and the other host are configured correctly.

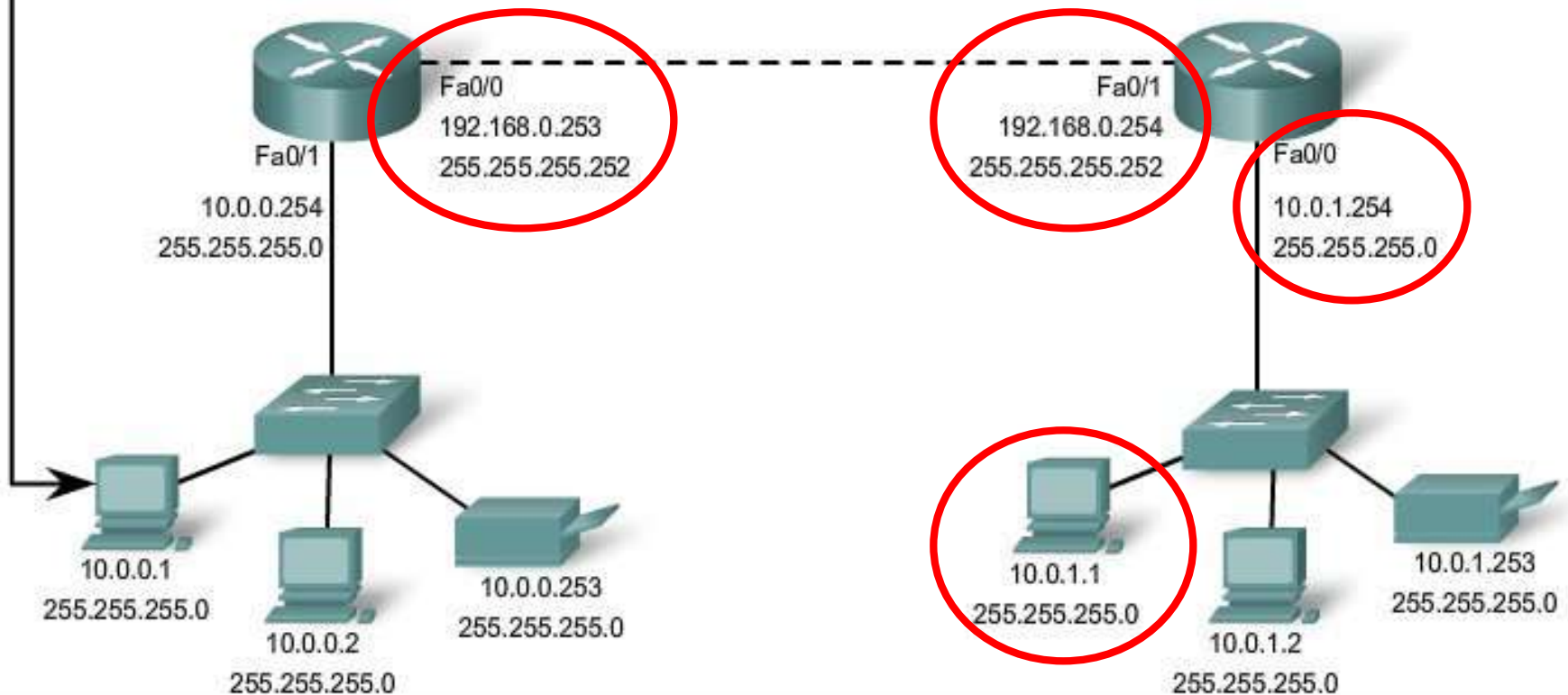
Testing Gateway Connectivity



A successful ping verifies both the host and gateway configurations.

Testing Remote Connectivity

```
C:\>ping 192.168.0.253  
C:\>ping 192.168.0.254  
C:\>ping 10.0.1.254  
C:\>ping 10.0.1.1
```



Ping each hop between the local and remote host.

Tracing and Interpreting Trace Results

C:\ping 10.1.0.2

Pinging 10.1.0.2 with 32 bytes of data:

Request timed out.

Request timed out.

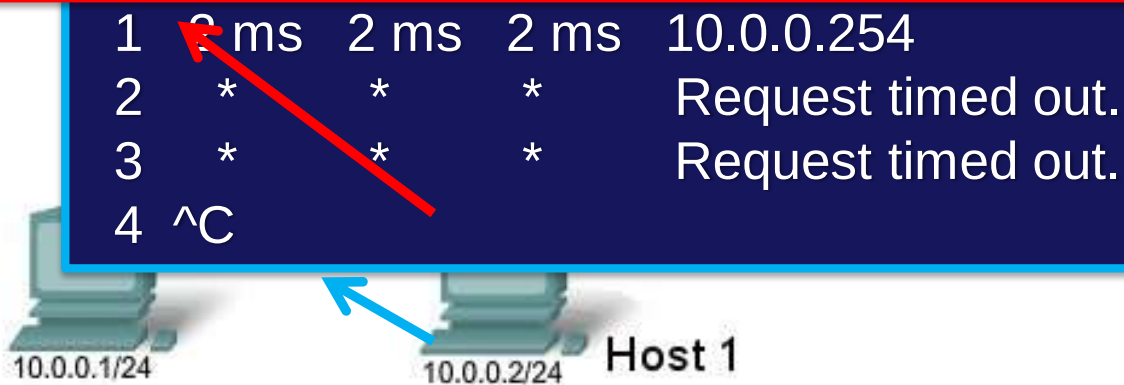
Request timed out.

Request timed out.

Ping statistics for 10.1.0.2:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)

The ping test failed.



1	2 ms	2 ms	2 ms	10.0.0.254
2	*	*	*	Request timed out.
3	*	*	*	Request timed out.
4	^C			

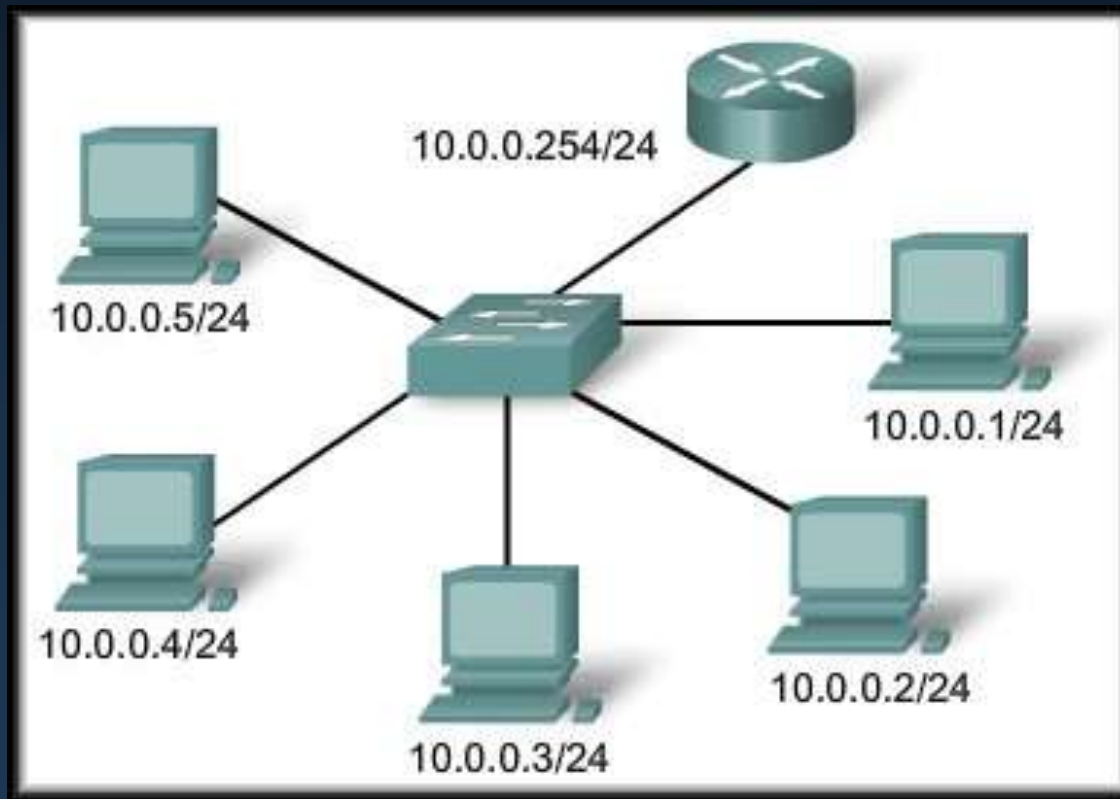
at this point.

10.1.0.254/24

0 hops

Configuring and Testing Your Network

Monitoring and Documenting Networks



Basic Network Baselines

- **Network Baseline:**
 - Process for studying the network at regular intervals.
 - Ensure that the network is working as designed.
 - More than a single report.
 - Health of the network at a certain point in time.
 - Accomplished over a period of time.
 - Varying times and loads will assist in creating a better picture of overall network performance.
 - Copy and paste the results from an executed **ping, trace, or other relevant command** into a text file.
 - Compare the results over time.

Basic Network Baselines

Run the
same test.

At different
times.

Compare
the results.

FEB 2, 2007 08:14:43

C:\>ping 10.66.254.159

Pinging 10.66.254.159 with 32 bytes of data:

```
Reply from 10.66.254.159: bytes=32 time<1ms TTL=128
Reply from 10.66.254.159: bytes=32 time<1ms TTL=128
Reply from 10.66.254.159: bytes=32 time<1ms TTL=128
Reply from 10.66.254.159: bytes=32 time<1ms TTL=128
```

MAR 17, 2007 14:41:06

C:\>ping 10.66.254.159

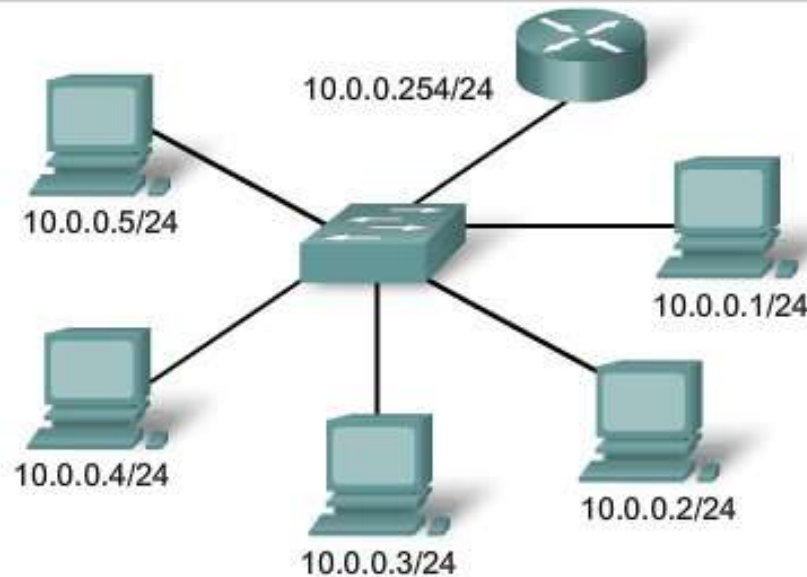
Pinging 10.66.254.159 with 32 bytes of data:

```
Reply from 10.66.254.159: bytes=32 time<6ms TTL=128
Reply from 10.66.254.159: bytes=32 time<6ms TTL=128
Reply from 10.66.254.159: bytes=32 time<6ms TTL=128
Reply from 10.66.254.159: bytes=32 time<6ms TTL=128
```

Basic Network Baselines

- Network Baseline:
 - Results from a command can be captured and saved as a text file for future study.
 - PING
 - TRACEROUTE
 - CISCO IOS

Learning About the Nodes on the Network

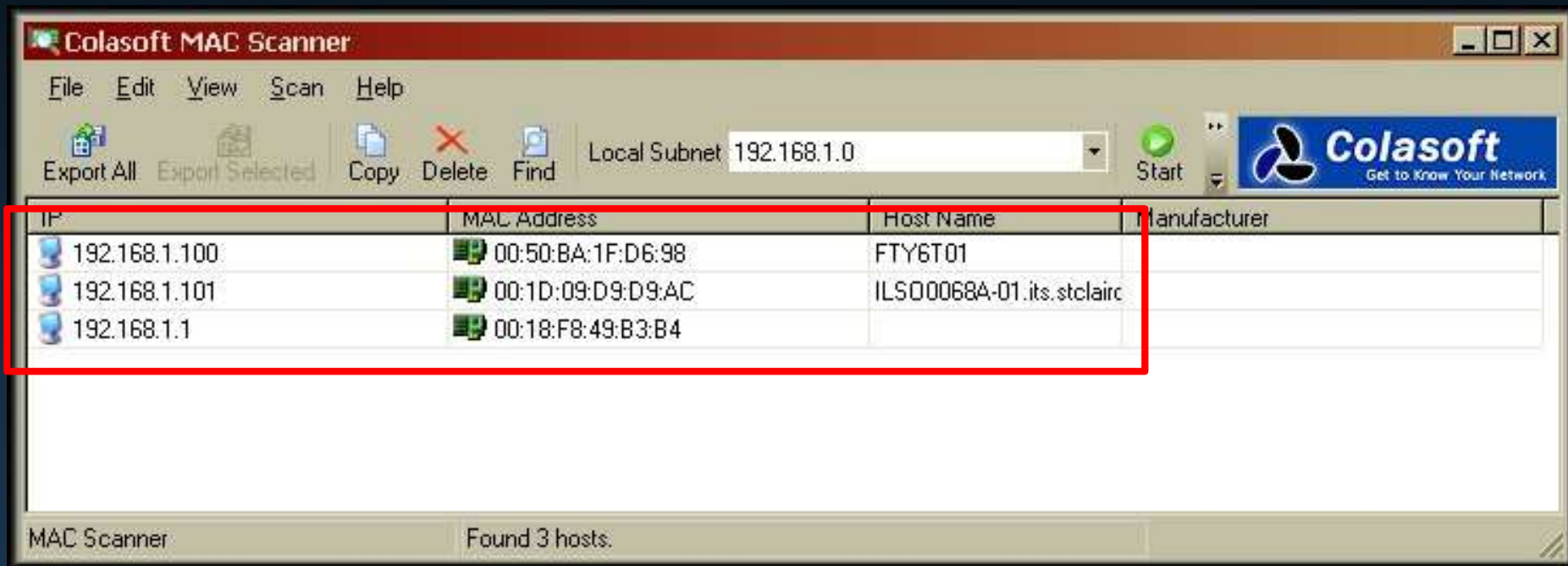


```
C:\>arp -a
```

Internet Address	Physical Address	Type
10.0.0.2	00-08-a3-b6-ce-04	dynamic
10.0.0.3	00-0d-56-09-fb-d1	dynamic
10.0.0.4	00-12-3f-d4-6d-1b	dynamic
10.0.0.254	00-10-7b-e7-fa-ef	dynamic

IP- MAC Address Pair

Ping Sweep / MAC Scanner Tools



Switch Connections

```
Sw1-2950#show mac-address-table
```

Mac Address Table

Vlan	Mac Address	Type	Ports
All	0014.a8a8.8780	STATIC	CPU
All	0100.0ccc.cccc	STATIC	CPU
All	0100.0ccc.cccd	STATIC	CPU
All	0100.0cdd.dddd	STATIC	CPU
1	0001.e640.3b4b	DYNAMIC	Fa0/23
1	0002.fde1.6acb	DYNAMIC	Fa0/14
1	0006.5b88.dfc4	DYNAMIC	Gi0/2
1	0006.5bdd.6fee	DYNAMIC	Fa0/23
1	0006.5bdd.7035	DYNAMIC	Fa0/23
1	0006.5bdd.72fd	DYNAMIC	Fa0/23
1	0006.5bdd.73b0	DYNAMIC	Fa0/23
1	000e.0cb6.2b51	DYNAMIC	Fa0/2
1	000f.8f28.b7b5	DYNAMIC	Fa0/18
1	0011.1165.8acf	DYNAMIC	Fa0/1
1	0013.720b.40c3	DYNAMIC	Fa0/19
1	0080.9120.1766	DYNAMIC	Fa0/8
1	00a0.c949.702a	DYNAMIC	Fa0/15
1	00c0.b770.6c19	DYNAMIC	Fa0/22
1	00c0.b770.6c8e	DYNAMIC	Fa0/21
1	00c0.b770.6c8f	DYNAMIC	Fa0/20
1	00e0.1e68.0987	DYNAMIC	Fa0/17

Multiple devices connected to Fa0/23